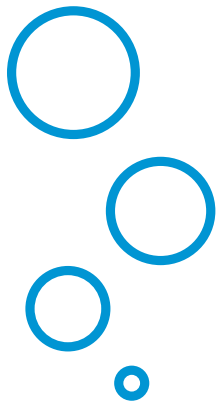


Brewing

Professional AST Guidebook (中文)



Brewing Professional

AST Guidebook V1.0 (中文/Simplified Chinese)

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致謝：

過去十多年來，無數的咖啡專業人士和具有天賦的教育工作者為 SCA 認證教育計畫自願付出了他們的寶貴的時間和專業知識。以下列出最新的貢獻者，他們藉由貢獻以及驗證目前的證書課程教材來延續此傳承。

Simon James, Genovese Coffee

Joost Leopold, De Koffieschool

Kyuho Choi, Principle Coffee Company

Brady Butler, Trail Blaze Coffee

Anita Bertok, Warda Coffee Roasters

Lara Hunter-Rodwell, Bewley's Ireland (former)

Tim Sturk, Cherry Coffee Training

Irina Blank

Alessandro Galtieri, Independent AST

Ole Sønstebo, Norsk kaffeinformasjon: Kaffe.no

Ellen Goormans, Cuperus Koffie (former)

Giorgos Konstantinopoulos, Coffee Island

注意：本指南替代了以前的课程文件。本指南仅适用于Brewing 许可的授权培训师。请不要与其他AST 或学习者共享此文档。 *Note: This guidebook replaces previous curriculum documents. This guidebook is only available to authorized trainers licensed in Brewing. Please do not share this document with other ASTs or learners.*

1. 一般信息 General Information

课程信息 Course Information

课程长度：最少 21 小时 Course Length: Minimum 21 hours

通过以下方式花费的时间可以满足最低课程时间： *Minimum course hours are met through time spent in the following ways:*

- AST 可亲自或通过远程学习提供实时指导。 The AST delivering live instruction in-person or through distance learning.
- AST 进行非正式和正式评估。（注意：这包括学习者参加在线笔试所花费的时间。） The AST conducting informal and formal assessments. (Note: This does not include time spent by the learner taking the online written exam.)
- 学习者参与实践课程活动以强化基础知识。 The learner engaging in activities that introduce and/or reinforce the theoretical or practical course objectives

冲煮中级建议学生已上过冲煮基础和感官技能基础课程。 Prerequisites: Brewing Intermediate.

建议已有 3 到 6 个月的工作经验，会使用自动滴滤机和手动滤泡咖啡设备或其他符合当地文化习惯的冲煮设备。 3 to 6 months' work experience preparing brews with an automatic drip filter brewer and a manual filter brewer or other culturally common brewer is recommended.

笔试信息： Written Exam Information:

在线笔试总题数： 40（每项价值 1 分） Total Number of Questions on Online Written Exam: 40 (worth one point each)

在线笔试总时间： 60 分钟 Total Time Allowed for Online Written Exam: 60 minutes

及格分数（在线笔试）： 80% Passing Score (Online Written Exam): 80%

在笔试期间，学习者可以使用“比例冲煮控制图”和计算器。在线笔试时会提供一个在线计算器。 The learner must have access to the Ratio Brewing Control Chart and a calculator during the written exam. The online written exam has an online calculator.

操作考试信息： Practical Exam Information:

操作考试总时间： 190 分钟 Total Time Allowed for Practical Exam: 190 minutes

及格分数： 总分的 80% Passing Score: Overall score of 80%

2. 课程说明和更新 Course Description and Updates

描述 Description

“冲煮高级”课程以“冲煮中级”课程中介绍的概念和技能为基础。学习者将更深入，更科学地了解“最优冲煮”的基本要素，操作冲煮参数时会发生什么以及如何根据咖啡冲煮控制图的引导做调整。学习者将对水及其对冲煮的影响有更深入的了解，特别是利用理想参数，测量和处理的方法。专业咖啡师拥有的最强大的工具是分析能力以及处理和控制变量的能力。这些变量可以帮助咖啡师了解如何最好地解释信息，然后提供解决方案或意见，以改善其客户的咖啡质量和服务。实践考试会根据课程中进行的的不同活动来测试专业技能，而笔试则会测试上述技能所需的背景知识。The Brewing Professional course builds upon the concepts and skills introduced in the Brewing Intermediate course. Learners take a deeper and more scientific look at the Essential Elements of Good Brewing, what happens when brewing parameters are manipulated, and how to master navigation of the coffee brewing control chart. Learners will gain a deeper understanding of water and its impact on brewing, specifically utilizing the ideals of aim, measure, and treatment. The most powerful tools that a professional brewer possesses is an analytical mind and the ability to process and manipulate a multitude of changing variables. These variables help the brewer to understand how best to interpret the information and then offer a solution or opinion that will improve the coffee quality, service, and delivery for their clients. A written exam tests professional course knowledge while a practical exam tests the skills described above based on different working activities carried out during the course.

先前版本的课程更新 Curriculum Updates from Previous Version

冲煮高级课程已进行了重大更新和修订，以提供清晰度，改进的结构以及与 AST 可以用来支持其提供的培训的参考文献的链接。AST 应仔细阅读本节以及第 4 节中的课程目标，以确保其教材是最新的。在实践考试中已添加了详细的说明和评分，以使 AST 和学习者更加清晰地了解整个课程。The Brewing Professional curriculum has been significantly updated and revised to provide clarity, an improved structure, and clear links to reputable references that ASTs can use to support the training they provide. ASTs should review in detail this section as well as the actual curriculum in Section 4 to ensure their teaching materials are up to date. Detailed instructions and scoring have been added to the practical exam to provide clarity to the AST and the learner.

课程变更：Changes to Curriculum:

- 现在，“探索核心冲煮变量”（1.0）是“冲煮准则”部分。“Exploring Core Brewing Variables”（1.0）is now the “Brewing Guidelines” section.
- “冲煮的基本要素”（1.1）现在是“冲煮的 7 个要素”（3.03.00）“Essential Elements of Brewing”（1.1）is now “The 7 Essential Elements of Brewing”（3.03.00）
- “温度”（1.2）现在是“水温”（3.03.05）“Temperature”（1.2）is now “Water Temperature”（3.03.05）
- “冲煮过程”部分中的“润湿（闷蒸）”（1.3）现在为“润湿”（3.04.03）。“Wetting (Blooming)”（1.3）is now “Wetting”（3.04.03）in the “Brewing Process” section.
- “搅动（扰流）”现在为“冲煮扰流”（3.03.06）“Agitation (Turbulence)” is now “Brew Turbulence”（3.03.06）
- “接触时间”（1.5）现在为“冲煮时间”（3.03.04）“Contact Time”（1.5）is now “Brewing Time”（3.03.04）

- “过滤材料” (1.6) 现在是“过滤介质” (3.03.08) “Filtering Materials” (1.6) is now “Filter Media” (3.03.08)
- “过滤器形状” (1.7) 现在是“重力过滤器形状” (3.02.02) , 并且是“冲煮方法和设备”栏目的一部分“Filter Shape (1.7)” is now “Gravity Filter Shape” (3.02.02), and is part of the “Brewing Methods & Equipment” section
- “萃取” (2.0) 现在是“溶出效率 (萃取)” (3.05.03) “Extraction” (2.0) is now “Solubles Yield (Extraction)” (3.05.03)
- “强度” (3.0) 现在是“溶解物浓度 (强度)” (3.05.04) “Strength” (3.0) is now “Solubles Concentration (Strength)” (3.05.04)
- “研磨尺寸” (4.1) 现在是“研磨设置–粒径分布” (3.03.03) “Grind Size” (4.1) is now “Grind Setting – Particle Size Distribution” (3.03.03)
- “烘焙的影响” (5.1) 现在是“烘焙程度” (3.01.02.02) , 并且是“咖啡知识”栏目的一部分。“The Effects of Roast” (5.1) is now “Roast Level” (3.01.02.02) and is part of the “Coffee Knowledge” section.
- “使用旁路” (6.1) 现在是“旁路” (3.07) “Using Bypass” (6.1) is now “Bypass” (3.07)
- “了解水” (7.0) 现在是“水质” (3.03.07) “Know Your Water” (7.0) is now “Water Quality” (3.03.07)
- “水的三个关键质量” (7.1) 现在是“水的起源” (3.03.07.01) 和“建议与要求” (3.03.07.02) “The Three Key Qualities of Water” (7.1) is now “Water Origins” (3.03.07.01), and “Recommendations & Requirements” (3.03.07.02)
- “碱度读数” (7.2) 现在是“碱度” (3.03.07.03) “Alkalinity Measure” (7.2) is now “Alkalinity” (3.03.07.03)
- “总硬度” (7.3) 现在是“总硬度” (3.03.07.04) “Total Hardness” (7.3) is now “Total Hardness” (3.03.07.04)
- “总溶解固体 (TDS)” 现在是“电导率” (3.03.07.05) “Total Dissolved Solids (TDS)” is now “Electrical Conductivity” (3.03.07.05)
- “水测试” (7.7) 现在是“水分析” (3.03.07.07) “Water Testing” (7.7) is now “Water Analysis” (3.03.07.07)
- “使用不同水冲煮” (7.8) 现在是“验证感官结果” (3.03.07.08) “Brewing Using Variable Waters” (7.8) is now “Verify Sensory Outcomes” (3.03.07.08)

课程删除: Deletions to Curriculum:

- 没有 None

操作考试的变更 Changes to the Practical Exam

- 学习者应单独完成每个任务, 而无需与其他学习者协商。 Learners should complete each task individually, without consultation with other Learners.

其他增补和修订 Other Additions and Revisions

- 本课程中描述的活动是强制性的, 旨在满足每个主题的目标, 并为每个学习者提供充分的实践考试准备。 The activities described in this curriculum are compulsory and are designed to meet the objectives of each topic, and to provide each learner with adequate preparation for the practical exam.
- 额外的练习时间由 AST 决定。 Additional practice time is at the discretion of the AST.

3. 按主题分布的书面笔试题 Written Exam Question Distribution by Topic

下表列出了有关在线考试问题的关键信息。

题库：这收录了每个**主题**可以在在线考试期间向学习者提出的问题。

提出的问题：这是学习者将在在线考试期间针对**每个主题**随机接收的问题数。该数字由课程设计者决定，目的是确保对课程的每个栏目和主题进行适当加权。

栏目权重：每个栏目标题旁边是该栏目中的问题所代表的全部考试的百分比。

在笔试或实践考试中，有些主题/目的不考。培训师仍应教授这些主题/目标。

The chart below sets forth key information regarding the online exam questions.

Question Pool: This is the number of questions **per topic** that are available to present to the learner during the online exam.

Questions Presented: This is the number of questions a learner will randomly receive **per topic** during the online exam. This number was determined by the creators for the purpose of ensuring that each section and topic of the course is weighted appropriately.

Section Weighting: Next to each section title is the percentage of the total exam represented by the questions in that section.

There are some topics/objectives that are not tested during the written or practical exam. The trainer should still address these topics/objectives.

考试栏目和主题 Exam Sections & Topics	题库 Question Pool	提出的问题 Questions Presented	考试栏目和主题 Exam Sections & Topics	题库 Question Pool	提出的问题 Questions Presented
3.01 栏目 咖啡知识 5% Section COFFEE KNOWLEDGE 5%			子主题：调整方法 Sub-Topic: Treatment Methods	3	2
1 主题：历史 Topic: History	中级授课 Taught in Intermediate		子主题：水质测试 Sub-Topic: Water Testing		活动 Activity
2 主题：烘焙程度 Topic: Roast Level	4	2	子主题：验证感官结果 Sub-Topic: Verifying Sensory Outcomes		活动 Activity
	活动 Activity			4	1
3.02 栏目 冲煮方法和设备 7.5% Section BREWING METHODS AND EQUIPMENT 7.5%			8 主题：过滤介质 Topic: Filter Media		活动/实践考试 Activity / Practical Exam
1 主题：刀盘类型 Topic: Grinder Burr Types	中级授课 Taught in Intermediate		3.04 栏目 冲煮工艺 7.5% Section BREWING PROCESS 7.5%		
2 主题：重力过滤器形状 Topic: Gravity Filter Shape			1 主题：冲煮设备的使用-重力冲煮器 Topic: Brewing Device Usage – Gravity Brewers	中级授课 Taught in Intermediate	
子主题：建议咖啡床深度 Sub-Topic: Recommended Bed Depth	2	1	2 主题：冲煮工艺 Topic: Brewing Processes	中级授课 Taught in Intermediate	
子主题：设备形状对咖啡床深的影响 Sub-Topic: Effect of Device Shape on Bed Depth	4	2	3 主题：润湿（闷蒸） Topic: Wetting (Blooming)		
子主题：设备形状对冲煮成品的影响 Sub-Topic: Effect of Device Shape on the Finished Brew	活动 Activity		子主题：水量 Sub-Topic: Water Quantity	2	1
3.03 栏目 冲煮指南 Section BREWING GUIDELINES 57.5%			子主题：润湿性 Sub-Topic: Wettability		授课，不考 Taught, not tested

1 主题：冲煮的 7 个基本要素 Topic: 7 Essential Elements of Brewing	4	2	子主题：基于保水率计算萃取率 Sub-Topic: Calculating Yield Based on Water Retention	2	1
2 主题：咖啡与水的比例 Topic: Coffee to Water Ratio	活动/实践考试 Taught in Intermediate		子主题：润湿阶段与成品冲煮的关系 Sub-Topic: Wetting Phase and Finished Brew Relationship	3	1
3 主题：研磨设置–粒径分布 Topic: Grind Setting – Particle Size Distribution			子主题：润湿应用 Sub-Topic: Wetting Application	活动 Activity	
子主题：粒径范围 Sub-Topic: Particle size ranges	2	1	3.05 栏目 冲煮分析 10% Section BREW ANALYSIS 10%		
子主题：影响粒径分布的因素 Sub-Topic: Factors Influencing Particle Size Distribution	2	1	1 主题：描述冲煮 Topic: Describe the Brew	活动/实践考试 Taught in 活动/实践考试 Intermediate	
子主题：测量粒径分布的方法 Sub-Topic: Methods for Measuring Particle Size Distribution	授课，不考 Taught, not tested		2 主题：平衡冲煮 Topic: Balanced Brew	活动/实践考试 Taught in Intermediate	
子主题：粒径分布对冲煮的影响。 Sub-Topic: Effect of Particle Size Distribution on The Brew.	活动 Activity		3 主题：溶解物萃取率（萃取） Topic: Solubles Yield (Extraction)	4	2
4 主题：冲煮时间 Topic: Brewing Time	3	2		活动/实践考试 Activity / Practical Exam	
	活动 Activity				
5 主题：水温 Topic: Water Temperature			4 主题：浓缩物浓度（强度） Topic: Solubles Concentration (Strength)	4	2
子主题：对不同可溶性化合物萃取率的影响 Sub-Topic: Effect on Extraction Rates of Different Soluble Compounds	4	2		活动/实践考试 Activity/ Practical Exam	
子主题：对煮好的咖啡的感官方面的影响 Sub-Topic: Effect on Sensory Aspects of Brewed Coffee	活动 Activity		3.06 栏目 保养 2.5% Section MAINTENANCE 2.5%		
6 主题：冲煮扰流 Topic: Brew Turbulence	4	2	1 主题：刀盘 Topic: Grinder Burrs		
	活动 Activity		子主题：刀盘边缘边质量 Sub-Topic: Burrs Edge Quality	活动 Activity	
7 主题：水质 Topic: Water Quality			子主题：更换要求 Sub-主题 Topic: Replacement Requirement	2	1
子主题：水的起源 Sub-Topic: Water Origins	2	1	3.07 栏目 旁路 10% Section BYPASS 10%		
子主题：建议与要求 Sub-Topic: Recommendations & Requirements	4	2	1 主题：使用旁路的好处 Topic: Benefits of Using Bypass	3	2
子主题：碱度 Sub-Topic: Alkalinity	4	3	2 主题：计算，测量和使用旁路 Topic: Calculate, Measure, and Chart Bypass	3	2
子主题：总硬度（TH） Sub-Topic: Total Hardness (TH)	5	3	3 主题：旁路对冲煮咖啡感官方面的影响 Topic: Effect of Bypass on The Sensory Aspects of Brewed Coffee	活动 Activity	
子主题：电导率 Sub-Topic: Electrical Conductivity	4	1	问题总数 Total Number of Questions	78	40

4. 搭配相应在线笔试问题的课程 Course Curriculum with Corresponding Online Written Exam Questions

该课程的课程设置如下，分为栏目，主题，子主题和目标。在课程的某些区域，创作者可能已经修改了课程，以创建更合乎逻辑，更适合层次的结构。修订记录在 2. *课程说明和更新*。处可以找到。

在线笔试题作为对特定目标的评估。这些问题已根据主题分组。主题内的所有问题均被视为主题“池” – 题库。从该库中，将随机选择一定数量的问题，并将其呈现给学习者。这是由于该主题中问题的随机性，如果特定主题的目标不止一个，则有可能无法针对该主题的所有目标对学习者进行测试。课程中还包括 AST 的详细说明，这些说明有助于说明内容以及如何实现目标。

- **重要的提示：在考试期间，学习者将需要“比率控制图 101017”。AST 请在考试前将此副本提供给学习者。**

The course curriculum is set forth below and is divided into Sections, Topics, Sub-Topics, and Objectives. In some areas of the curriculum, the creators may have revised the curriculum in order to create a more logical, level-appropriate structure. Revisions are noted in 2. *Course Description and Updates*.

All online written exam questions were developed as an assessment for a specific objective. The questions have been grouped according to topic. All questions within a topic are considered the topic “pool.” From this pool, a certain number of questions will be randomly selected and presented to the learner. If a particular topic has more than one objective, there is a possibility that the learner will not be tested on all objectives in the topic. This is due to the randomization of the questions from that topic. Also included in the curriculum are detailed notes for the ASTs that help explain the content and how to achieve the objectives.

- **IMPORTANT NOTE: Learners will need the Ratio Control Chart 101017 during the exam. ASTs please provide a copy of this to the learners prior to the exam.**

3.01 |栏目|咖啡知识| COFFEE KNOWLEDGE

1 主题

主题 Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考文献 Reference
3.01.01 历史 History	该主题是在中级水平上教授和测试的 This topic was taught and tested at Intermediate level.			
3.01.02 烘焙程度 Roast Level	1. 识别不同烘焙对可溶物萃取率，咖啡的感官方面以及各种冲煮设备的性能的影响。Identify the impact different roasts have on solubles yield, sensory aspects of brewed coffee, and performance in various brewing devices	咖啡熟豆的溶解度受许多因素影响，包括烘焙颜色和烘焙方法。专业的咖啡师必须意识到，当所有冲煮参数保持相等时，烘焙成不同颜色和/或采用不同方法烘焙的咖啡将产生不同的结果。The solubility of roasted coffee is influenced by many factors including roast color, and roast methodology. The professional brewer must be aware that coffees roasted to different colors, and/or with different methodologies will produce different results when all brewing parameters remain equal.	问题编码 Question ID 100008257586 深烘焙如何影响冲煮？ How does a dark roast profile affect the brewing process? 深烘焙中较高的油脂含量在萃取过程中释放出更多的风味。 Higher oil content of dark roast releases more flavor during extraction. 深焙咖啡豆的组织较脆，研磨时会产生较多的细粉，而且较深焙的咖啡可溶物质的溶出率（萃取）也较快。 <u>A darker roast is more brittle, yielding more fines upon grinding and darker roasted soluble matter dissolves (extracts) quicker.</u> 深焙咖啡豆因为豆子重量较轻，所以同等重量下咖啡豆的数量较多，因此较多的咖啡豆数量等同于较多的萃取。 A darker roast has more beans by volume because the beans are lighter in weight, therefore more beans equals more extraction. 烘焙程度不会影响咖啡的溶解度。 Roast level does not affect the solubility of coffee.	SCAA & Lingle, The Coffee Brewing Handbook, 25 Illy & Viani, Espresso Coffee - The Science of Quality, 193 Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 261; 264; 323

			问题编码 Question ID 100008257588 以下哪个说法是正确的？ Which statement below is correct? <u>不同的烘焙曲线会影响咖啡豆的密度，这可能会导致某些冲煮方法的流速发生变化。</u> <u>Different roast profiles will affect the density of the beans, and this may cause changes in flow rate with some brewing methods.</u> 深咖啡熟豆在滴滤器中始终会产生较慢的流速。 Dark roasted coffee will always produce a slower flow rate in drip filter devices. 不同的烘焙曲线会产生不同感官品质的滴滤式咖啡，但不会影响冲煮时的流速。 Different roast profiles will produce different sensory qualities in drip filter coffee but will not affect flow rate. 不同的烘焙曲线会产生不同感官品质的咖啡，但不会影响溶解度。 Different roast profiles will produce different sensory qualities in brewed coffee, but will not affect solubility.	
			问题编码 Question ID 100008257589 以下哪个说法是正确的？ Which statement below is correct? <u>较深焙的咖啡熟豆通常比浅焙的咖啡熟豆密度低，并且具有更多的 CO₂，这可能会影响吸水性和膨胀度。</u> <u>Darker roasted coffee is typically less dense and has more CO₂ than lighter roasted coffee, and this may affect the wettability and bloom.</u> 较深焙的咖啡熟豆通常比浅焙的咖啡熟豆密度低，并且具有较少的 CO₂，这将降低咖啡粉的吸水性。 Darker roasted coffee is typically less dense, and has less CO₂ than lighter roasted coffee, and this will always decrease the wettability of the coffee grounds. 烘焙曲线会影响咖啡的感官品质，但不会影响咖啡粉的吸水性。 The roast profile will affect sensory qualities of brewed coffee but will not affect wettability of the coffee grounds. 较浅焙的咖啡熟豆通常比深焙的咖啡熟豆密度更大，且具有更多的 CO₂，但这不会影响吸水性或膨胀度。 Lighter roasted coffee is typically more dense and has more CO₂ than darker roasted coffee, but this will not affect the wettability or the bloom.	

		问题编码 Question ID 100008257590 以下哪种说法是正确的? Which statement below is correct?? <u>咖啡熟豆的溶解度受烘焙方法的影响, 包括颜色, 发展程度和烘焙速度。</u> <u>Solubility of roasted coffee is affected by roasting methodology, including color, development, and speed of roast.</u> 咖啡的溶解度仅受烘焙颜色的影响。 Solubility of roasted coffee is only affected by roast color. 咖啡的溶解度和感官品质仅受烘焙颜色的影响。 Solubility and sensory qualities of roasted coffee are only affected by roast color. 烘焙方法会影响咖啡的感官品质, 但不会影响溶解度。 The roast profile will affect sensory qualities of brewed coffee but will not affect solubility.	
	2. 探索可优化或改善浅烘焙咖啡冲煮的技术 Explore techniques that optimize or improve brews of lighter roasted coffee	该主题有一个活动。 This topic has an activity. 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide	
	3. 探索优化或改善深烘焙咖啡冲煮的技术 Explore techniques that optimize or improve brews of darker roasted coffee		

3.02 |栏目|冲煮方法和设备 BREWING METHODS & EQUIPMENT

1 主题

主题 Topic	子主题 Sub-Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考文献 Reference
3.02.01 刀盘类型 Grinder Burr Types	该主题是在中级水平上教授和测试的 This topic was taught and tested at Intermediate level.				
重力过滤器 形状 3.02.02 Gravity Filter Shape	3.02.02.01 建议咖啡床深度 Recommended Bed Depth	1. 识别重力过滤冲煮设备的建议咖啡床深度 Identify the recommended bed depth for gravity filter brewing devices		问题编码 Question ID 100008257591 为了产生更均匀的萃取，重力冲煮设备的建议咖啡床深度范围是多少？ To produce a more even extraction, what is the recommended coffee bed depth range for gravity brewing devices? 2.5 厘米 - 5 厘米 (1 英寸-2 英寸) 2.5cm - 5cm (1in - 2in) 1.25 厘米 - 2.5 厘米 (0.5 英寸-1 英寸) 1.25cm - 2.5cm (0.5in - 1in) 2 厘米-10 厘米 (0.75 英寸-4 英寸) 2cm - 10cm (0.75in - 4in) 6.5 厘米-12.7 厘米 (2.5 英寸-5 英寸) 6.5cm - 12.7cm (2.5in - 5in)	SCAA & Lingle, The Coffee Brewing Handbook, 34, 41-43
				问题编码 Question ID 100008257592 为了减少在重力过滤冲煮设备中通道效应的风险，建议的咖啡床深度范围是多少？ To reduce the risk of channelling in a gravity filter brewer, what is the recommended coffee bed depth range? 2.5 厘米 - 5 厘米 (1 英寸-2 英寸) 2.5cm - 5cm (1in - 2in) 1.25 厘米 - 2.5 厘米 (0.5 英寸-1 英寸) 1.25cm - 2.5cm (0.5in - 1in) 2 厘米-10 厘米 (0.75 英寸-4 英寸) 2cm - 10cm (0.75in - 4in) 6.5 厘米-12.7 厘米 (2.5 英寸-5 英寸) 6.5cm - 12.7cm (2.5in - 5in)	

	3.02.02.02 设备形状对咖啡床深的影响 Effect of Device Shape on Bed Depth	1. 识别重力冲煮设备的形状将如何影响咖啡床层深度 Identify how the shape of a gravity brewing device will affect the bed depth		问题编码 Question ID 100008257593 当选择一个重力冲煮设备，要确定其过滤器/冲煮篮的大小可以适用于即将研磨咖啡的量，下列因素是最相关？ When choosing which filter/brew basket size of a gravity brewing device to use for an amount of ground coffee, which of the following factors is most relevant? 研磨设定 Grind setting 水温 Water temperature 产生的咖啡床深度 Resulting coffee bed depth 冲煮时间 Brewing time 问题编码 Question ID 100008257594 过滤器的尺寸和形状在滴滤式冲煮中有什么关键影响？ How is filter size and shape relevant to drip filter brewing? <u>咖啡床的深度取决于过滤器的尺寸和形状，需要适当的咖啡床层深度以确保均匀冲煮和合适的停留时间。</u> <u>The coffee bed depth is determined by the filter size and shape. A suitable bed depth is required for uniform brewing and suitable dwell time.</u> 水始终会更均匀地流经平底过滤器，较大的过滤器始终可以产生更高质量的冲煮冲煮。Water always flows more evenly through a flat bottom filter. Larger filters always produce better quality brews. 水始终会更均匀地流过锥形过滤器，较小的过滤器始终可以产生更高质量的冲煮冲煮。Water always flows more evenly through a cone shaped filter. Smaller filters always produce better quality brews. 与大的平底过滤器相比，小锥形或梯形过滤器更适合水洗咖啡。Small cone or truncated cone filters are better suited to washed processed coffees, than large flat bottom filters.	SCAA & Lingle, The Coffee Brewing Handbook, 34, 41- 43 Rao, <i>Everything but Espresso</i>, Chapter 5
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			问题编码 Question ID 100008257595 过滤器的形状很重要，因为它会影响咖啡床的形状（例如：圆锥形，梯形，平底形）。为什么我们要选择一种形状而不是另一种形状？ The shape of the filter is important as it affects the shape of the coffee bed (e.g. cone, truncated cone, flat). Why might we choose one shape over the other? 拥有合适的咖啡床深度 To have a suitable coffee bed depth 以匹配咖啡与水的比例 To match coffee to water ratio 以匹配所选的研磨尺寸 To match with the grind size chosen 以获得正确的饮料澄清度 To obtain the correct rate of clarification of the beverage	
			问题编码 Question ID 100008257596 选择两 (2) 个最合适的答案，以使过滤器形状与冲煮量匹配： Choose the two (2) most suitable answers regarding matching filter shape to brew volume: 冲煮量大于 500ml 的时候，建议使用平底过滤器，因为它更可能产生合适的咖啡床深度； A flat bottom filter is recommended with brews of more than 500mls because it is more likely to produce a suitable coffee bed depth. 冲煮量大于 500mls 的时候，建议使用锥形过滤器，因为它更可能产生合适的咖啡床深度。 A cone filter is recommended with brews of more than 500mls because it is more likely to produce a suitable coffee bed depth. 咖啡床的深度仅受研磨咖啡剂量的影响，而不受冲煮量或过滤器形状的影响。 The coffee bed depth is only affected by the ground coffee dose and is not affected by brewing volume or filter shape. 咖啡床的深度受过滤器形状（圆锥形/平底形/梯形）的影响，因此建议过滤器的形状适合冲煮量。 The coffee bed depth is affected by the shape of the filter (cone/flat bottom/truncated cone), so it is recommended that the filter shape is appropriate for the brewing volume.	

	3.02.02.03 设备形状对冲煮成品的影响 Effect of Device Shape on the Finished Brew	1. 将相同的冲煮计划应用于所有 3 种设备时，比较并对比使用 3 种常见重力过滤设备形状（圆锥形，圆锥台形，平底）冲煮的咖啡的感官体验 然后 - 使用针对每个冲煮设备优化冲煮计划，对比感官体验 Compare and contrast the sensory aspects of coffee brewed using 3 common gravity filter device shapes (cone, truncated cone, flat bottom) when applying the same brew plan to all 3 devices Then - Compare and contrast the sensory aspects using brew plans optimized for each of the brewing devices	该主题有一个活动。 This topic has an activity. 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide	
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3.03 |栏目|冲煮指南 BREWING GUIDELINES

7 主题

主题 Topic	子主题 Sub-Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考文献 Reference
3.03.01 冲煮的 7 个基本要素 The 7 Essential Elements of Brewing		1. 确定冲煮的 7 个基本要素。 2. 了解冲煮的 7 个基本要素中的每一个如何相互影响 3. 意识到技术的一致应用对于控制和产生一致的结果至关重要 1. Identify each of the 7 essential elements of brewing 2. Aware of how each of the 7 essential elements of brewing may interact with each other 3. Aware that consistent application of technique is essential for controlling and producing consistent results	专业咖啡师必须了解冲煮的 7 个基本要素中的每个要素如何用于改变冲煮的感官特性，这一点至关重要。为了掌握这些要素，专业的咖啡师还必须意识到这些要素之间如何相互作用（例如，在滴滤式冲煮中更换过滤介质可能会改变接触时间）。 AST 应该培训学习者以分析这 7 种冲煮的基本要素以适用于任何设备或冲煮手法，并严格探	问题编码 Question ID 100008257597 如果在滴滤过滤器冲煮中使用滤纸代替金属（筛）过滤网，会影响结果吗？ If a paper filter is used instead of a metal (sieve) filter in a drip filter brewer, would it affect the result? 是，流速可能会改变，并且接触时间可能会受到影响。 Yes. The flow rate may change, and the contact time may be affected. 否，过滤式咖啡滤杯的所有过滤介质设计均会产生相同的结果。 No. All filter media for drip filter brewers are designed to produce the same results. 是，冲煮咖啡流经金属滤网的时间通常会更长一些。 Yes. It always takes more time for the brew to flow through a metal filter. 是，冲煮咖啡流经金属滤网的时间通常会缩短一些。 Yes. It always takes less time for the brew to flow through a metal filter.	SCAA & Lingle, <i>The Coffee Brewing Handbook</i> Rao, <i>Everything but Espresso</i> Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, Chapter 13 Engineeringtoolbox.com, <i>Water - Density, Specific Weight And Thermal Expansion Coefficient</i>. [online]

			索如何使用设备或技术制作优质饮料。 It is essential that the professional brewer understands how each of the 7 essential elements of brewing can be used to change the sensory characteristics of a brew. To develop mastery with these elements, the professional brewer must also be aware of how the elements interact with each other (e.g. changing the filter media in a drip filter brewer may change contact time). The AST should be training learners to analyze how the 7 essential elements of brewing apply to any device, or brewing technique, and rigorously explore how a device or technique can be used to produce quality beverages.	问题编码 Question ID 100008257598 使用滴滤机时如何增加接触时间？ How can contact time be increased when using a drip filter brewer? 更精细的研磨设置，增加闷蒸的时间，增加注水时间，使用不同的过滤介质，改变扰流的施加时间和方式。 Finer grind setting; Increase duration of pause for bloom; Increase water delivery duration; Use different filter media; Change when and how turbulence is applied 更粗的研磨设置，减少闷蒸的时间，增加注水时间和方式。 Coarser grind setting; Decrease duration of pause for bloom; Increase water delivery duration; Use different filter media; Change when and how turbulence is applied 只有通过细研磨才能增加接触时间。 Contact time can only be increased by grinding finer. 只能通过细研磨和/或增加注水时间来增加接触时间。 Contact time can only be increased by grinding finer and/or increasing water delivery duration.	
				问题编码 Question ID 100008257599 1 升水在 93°C 时的重量为： 1 Liter of water at 93°C weighs: 1 公斤 1 kilogram 963 克 963 grams 1.04 公斤 1.04 kilograms 996 克 996 grams	

			问题编码 Question ID 100008257600 在其他所有参数不变的情况下，在滴滤式冲煮机中更改咖啡与水的比例会影响咖啡的可溶物浓度。冲煮的其他哪些方面可能会受到影响？ Changing the coffee to water ratio in a drip filter brewer, with all other parameters unchanged, will affect the solubles concentration of a brew. What other aspects of the brew may be affected? 溶解量、萃取率、冲煮时间可能都会改变，咖啡的风味平衡性也会改变。Solubles yield may change; Rate of extraction may change; Contact time may change; Balance of taste may change. 溶解量不会改变，冲煮时间可能会改变，但咖啡的风味平衡性不会影响。Unchanged solubles yield; Contact time may change; Unchanged balance of taste. 只有可溶物浓度会改变。Only the soluble concentration will change. 只有接触时间会改变。Only the contact time will change.	
3.03.02 咖啡与水的比 Coffee to Water Ratio	该主题是在中级水平上教授和测试的 This topic was taught and tested at Intermediate level.			

3.03.03 研磨设置-粒径分布 Grind Setting - Particle Size Distribution	3.03.03.01 粒径范围 Particle Size Ranges	1. 查看与常规冲煮设备配合使用的大致粒径范围 Review the approximate particle size ranges for use with common brewing devices	粒径分布被认为是生产高质量冲煮咖啡和选择磨豆机的重要因素。 AST 不应在此活动中，将一个磨豆机与另一个磨豆机比较高下。 理想的结果是让学习者得出结论，即粒径分布确实会影响冲煮过程和感官特性并描述差异。 可以并且应该进行首选讨论，但是还没有一个普遍的“正确答案”，对于该问题，粒径分布被认为是最佳的。 Particle size distribution is often discussed as an important factor in producing high-quality brewed coffee and selecting grinders. The AST should not use this activity to promote one grinder over another.	问题编码 Question ID 100008257602 将冲煮设备与典型的平均咖啡粉粒径匹配。 Match the brewing device with the typical average ground coffee particle sizes. - 土耳其壶 Ibrik coffee - 摩卡壶 Moka pot - 重力滴滤器 Gravity filter - 浸泡式冷萃设备 Immersion cold brew 400 - 800 微米 400 - 800 microns 850 - 1000 微米 850 - 1000 microns - 约 100 微米 Around 100 microns 250 - 350 微米 250 - 350 microns A = 3, B = 4, C = 1, D = 2	Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i> , 311 - 316
				问题编码 Question ID 100008257603 研磨滴滤咖啡时的典型平均粒径是多少 What is the typical average particle size when grinding for drip filter coffee? 400-800 微米 400 - 800 microns 900-1000 微米 900 - 1000 microns 300-400 微米 300 - 400 microns 100-300 微米 100 - 300 microns	Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i> , 313
	3.03.03.02 影响粒径分布的因素 Factors Influencing Particle Size Distribution	1. 确定影响研磨熟豆粒径分布的因素 Identify factors that influence particle size distribution of ground roasted coffee	The desired outcome is for the learner to conclude that particle size distribution does impact the brewing process and sensory characteristics and describe the differences. Preference discussions can and should occur, but there is not a universal "right answer"	问题编码 Question ID 100008860197 以下哪些因素会影响咖啡粉的粒径分布？ Which of the following are factors that influence the particle size distribution of ground coffee? 脆性 Brittleness 烘焙程度 Roasting degree 熟豆的水分含量 Moisture content of the roasted bean 以上所有 All of the above	Illy & Viani, <i>Espresso Coffee - The Science of Quality</i>, 224 - 227 SCAA & Lingle, <i>The Coffee Brewing Handbook</i>, 25 & 27

			for which particle size distribution is considered best.	问题编码 Question ID 100008860198 哪种程度的烘焙通常会产生更细的粒径分布？ Which degree of roast will typically produce a particle size distribution with more fines? 深 Dark 浅 Light 中等 Medium 所有程度的烘焙都会导致一样的粒径分布 All roast degrees will give the same particle size distribution	Illy & Viani, <i>Espresso Coffee - The Science of Quality</i> , 224 - 227 SCAA & Lingle, <i>The Coffee Brewing Handbook</i> , 25 & 27
	3.03.03.03 测量粒径分布的方法 Methods for Measuring Particle Size Distribution	1. 意识到可以使用激光衍射粒径分析仪或筛子来确定磨豆机的粒径分布以及如何绘制结果图 Aware that laser diffraction particle size analyzers or sieves can be used for determining the particle size distribution of a grinder, and how the results are charted		授课，不考。 Taught, not tested.	Illy & Viani, <i>Espresso Coffee - The Science of Quality</i> , 223 - 224
	3.03.03.04 粒径分布对冲煮的影响。 Effect of Particle Size Distribution on the Brew.	1. 描述冲煮咖啡的感官方面与所用粒径范围之间的关系 Describe the relationship between sensory aspects of brewed coffee and the range of particle sizes used	这项活动提供了一个机会来探索粒径分布与冲煮咖啡的感官特征之间的关系并验证这一想法。 This activity provides an opportunity to explore the relationship between particle size distribution and sensory characteristics of brewed coffee and validate this idea.	该主题有一个活动。 This topic has an activity. 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide	

3.03.04 Brewing Time		1. 确定改变水接触时间以适合冲煮方法和研磨设置的方法 2. 分析接触时间对冲煮咖啡的感官的影响 3. 分析接触时间变化对可溶物萃取率和可溶物浓度的影响 1. Identify methods of varying water contact time to suit the brewing method and the grind setting 2. Analyze the effect variations of contact time have on the sensory aspects of brewed coffee 3. Analyze the effect variations of contact time have on solubles yield and solubles concentration	接触时间应与冲煮设备的合适研磨设置相对应。 当以合适的研磨设置对应冲煮时，接触时间的变化会影响可溶物的萃取率和可溶物浓度。但是，如何改变接触时间将取决于设备以及可用的控制措施，包括（但不限于）： -过滤的开始（例如，从虹吸管中移除热源） -注水时间（例如，流量变化，或使用间段性冲法） Contact time should correspond with a suitable grind setting for the brewing device. When brewing with coffee grounds at a suitable grind setting, variations in contact time will affect solubles yield and solubles concentration. However, how the contact time is varied will depend on the device, and the control measures available, including	问题编码 Question ID 100008257607 在滴滤机中，接触时间受以下哪些因素影响： In a drip filter brewer, the contact time will be affected by: 水流入冲煮室的速度 The rate that water flows into the brewing chamber 研磨设置 The grind setting 扰流的数量和类型 The amount and type of turbulence 以上所有 All of the above 问题编码 Question ID 100008257608 如果您增加冲煮的接触时间，通常 If you increase the contact time of a brew you will typically 萃取更多的可溶性物质 Extract more solubles 萃取更少的可溶性物质 Extract fewer solubles 减少苦味 Decrease bitterness 增加对酸度的感知 Increase the perception of acidity	SCAA & Lingle, <i>The Coffee Brewing Handbook</i>, 23 & 49 Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 365
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			(but not limited to): - Initiation of filtration (e.g. removing heat source from syphon) - Water delivery duration (e.g. flow rate variation, or use of pulsing) Note: The rate of extraction will typically decline as contact time progresses, due to less material being available for extraction and/or diffusion limiting extraction.	问题编码 Question ID 100008257609 冲煮周期的哪一部分萃取速度最快，可溶性最高？ Which part of the brewing cycle has the most rapid rate of extraction and yields the most solubles? 最后一部分 Final third 第一部分 First third 第二部分 Second third	
			该主题有一个活动。 This topic has an activity. 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide		
3.03.05 水温 Water Temperature	3.03.05.01 对不同可溶性化合物萃取率的影响 Effect on Extraction Rates of Different Soluble Compounds	1. 解释冲煮水温度与咖啡中不同化合物的萃取率之间的关系 Explain the relationship between brewing water temperature and the extraction rates of different compounds in coffee	尽管建议水温为 93°C (+/-3°C) /195°F-205°F，但应鼓励学习者在范围之外的温度下进行冲煮，因为在某些情况下，它可能有助于纠正冲煮不佳的现象。 Although water temperature of 93°C (+/-3°C)/195°F - 205°F is recommended, learners should be encouraged to explore brewing with temperatures outside this range, because in some circumstances it may assist in correcting an unsatisfactory brew.	问题编码 Question ID 100008257610 当用 85°C/185°F 的水冲煮时，最终的冲煮结果会有什么特性？ When brewing with 85°C/185°F water, what characteristics might be expected in the finished brew? 不平衡和苦涩 Unbalanced and bitter 不平衡和酸 Unbalanced and sour 不平衡和甜 Unbalanced and sweet 过强 Overly strong	SCAA & Lingle, <i>The Coffee Brewing Handbook</i>, 29-33 Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 365-369
				问题编码 Question ID 100008257611 当用 98°C/208°F 的水冲煮时，最终的冲煮结果会有什么特性？ When brewing with 98°C/208°F water, what characteristics might be expected in the finished brew? 不平衡和苦涩 Unbalanced and bitter 不平衡和酸 Unbalanced and sour 不平衡和甜 Unbalanced and sweet 不平衡和弱 Unbalanced and weak	

			问题编码 Question ID 100008257612 选择以下最能使句子完整的答案。冲煮水的温度低于 90°C/195°F_____。 Select the answer below that best completes the sentence. A brewing water temperature below 90°C/195°F _____. 切勿用于煮咖啡 Should never be used for brewing coffee 可能适合较深的烘焙和/或更长的接触时间 <u>May be suitable with darker roasts and/or longer contact time</u> 将始终产生淡淡的风味并萃取更多的单宁 Will always produce flat flavors and extract more tannins 不会产生 18%-22%的可溶物萃取率 Will not produce a solubles yield of 18% - 22%	
			问题编码 Question ID 100008257613 选择以下最能使句子完整的答案：使用滴滤过滤器冲煮器时，改变水温会_____。 Select the answer below that best completes the sentence: When using a drip filter brewer, changing water temperature from one brew to the next _____. 将仅影响可溶物的萃取率 Will only affect the solubles yield 将影响杯中的风味、味道和口感 Will only affect the profile of taste, flavor, and body 将可能影响可溶物的萃取率，和杯中的风味、味道和口感 <u>Will likely affect the solubles yield, and the profile of taste, flavor and body</u> 将仅影响流速，和杯中的风味、味道和口感 Will only affect the flow rate, and the profile of taste, flavor, and body	
3.03.05.02 对咖啡的感官方面的影响 Effect on Sensory Aspects of Brewed Coffee	1. 总结冲煮水温度和冲煮咖啡感官之间的关系 Summarize the relationship between brewing water temperature and the sensory aspects of brewed coffee	该主题有一个活动。 This topic has an activity. 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide		

3.03.06 冲煮扰流 Brew Turbulence		1. 识别扰流变化与成品冲煮差异之间的关系 Identify the relationship between turbulence variation and differences in the finished brew	扰流可以多种方式变化，包括（但不限于）： -类型（例如浇注，搅拌，煮沸） -持续时间 - 频率 -应用时（例如冲煮周期的开始，中间或结束） -力量 -速度 扰流的变化会影响冲煮的各个方面，包括（但不限于）： -溶出率 -可溶性浓度 - 味道 -流量 - 干净度 Turbulence can be varied in many ways, including (but not limited to): - Type (e.g. pouring, stirring, boiling) - Duration	问题编码 Question ID 100008257614 保持所有其他参数相同，增加的扰流如何影响冲煮？ Keeping all other parameters the same, how does increasing turbulence affect the brew? 萃取将增加 Extraction will increase 萃取将减少 Extraction will decrease 萃取将不太均匀 Extraction will be less uniform 味道将不平衡 Taste will be unbalanced 问题编码 Question ID 100008257616 使用自动咖啡滤泡机设定 320 秒的送水，下列哪一个冲煮循环可以产生最多的扰动，且提升最多的萃取率？（仅选择一个正确答案） When using a 320 second water delivery cycle in an automatic filter brewer, which of the following pulse cycles will create the most turbulence, and increase solubles yield the most? (Choose only ONE correct answer) 10 秒开启，10 秒关闭，16 次 10 Seconds on, 10 seconds off, 16 times 80 秒开启，80 秒关闭，2 次 80 Seconds on, 80 Seconds off, 2 times 320 秒开启，0 秒关闭 320 seconds on, 0 seconds off 20 秒开启，20 秒关闭，8 次 20 Seconds on, 20 seconds off, 8 times	SCAA & Lingle, <i>The Coffee Brewing Handbook</i>, 34-35 Rao, Everything but Espresso, 3-4 & 21- 26
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3.03.07 水质 Water Quality	3.03.07.01 水的起源 Water Origins	1. 讨论各种来源的冲煮水中矿物质和其他物质的来源 Discuss origins of minerals and other substances in brewing water from various sources	水的组成可能因位置，水源类型和市政处理而异。学习者应能够将高矿物质水与地下水源相匹配，将较低矿物质水与地表水源相匹配，将蒸馏或反渗透处理零矿水匹配，对市政处理采用含氯或氯胺的水相匹配。 Water composition may vary depending on location, source type, and municipal treatment. Learners should be able to match high mineral waters to underground sources, lower mineral water to surface sources, zero mineral water to distilled or reverse osmosis treatment, and water containing chlorine or chloramine to municipal treatment.	问题编码 Question ID 100008282080 将水源与典型的矿物质含量匹配。 Match the water source to the typical mineral content. a. 地下水来源 Underground water sources b. 地表水源 Surface water sources c. 蒸馏或反渗透 Distilled or reverse osmosis 1. 零/超低矿物质含量 Zero/ultra-low mineral content 2. 矿物质含量低 Low mineral content 3. 高矿物质含量 High mineral content a = 3, b = 2, c = 1 问题编码 Question ID 100008858046 选择以下最能使句子完整的答案：地下水源通常与_____有关。 Select the answer below that best completes the sentence: Underground water sources are typically associated with _____. 矿物质含量低 Low mineral content 零/超低矿物质含量 Zero/ultra-low mineral content 矿物质含量高 High mineral content	Wellinger & SCA, <i>SCA Water Quality Handbook</i>, Chapters 2 & 3 Wellinger & SCA, <i>The Water Chart</i>, 5 Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 383
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	3.03.07.02 建议与要求 Recommendations & Requirements	1. 确定要求，包括用于定制冲煮咖啡的水的三个主要措施： -总硬度-碱度 -pH Identify the requirements, including the three central measures used to characterize water for brewing coffee: - Total hardness - Alkalinity - pH	冲煮水的要求 还包括：无气味，卫生，钠含量低，不含铁，铅，氯和其他不合适或有毒的化合物。 注意：SCA 关于水成分的水质手册建议基于技术和感官考虑。建议以外的水成分可能会产生可接受的结果，但可能会导致冲煮设备的腐蚀或水垢。 注意：碱性，pH 和碱度之间的相似性经常使学习者感到困惑。AST 应该预料到这方面的问题，并准备解释差异。 Requirements of water brewing also include that it is: odor free, hygienic, low in sodium, and free of iron, lead, chlorine, and other unsuitable or toxic compounds. Note: The SCA Water Quality handbook recommendations for water composition are based on technical and sensory considerations. Water compositions outside the recommendations may	问题编码 Question ID 100008282081 选择以下最能使句子完整的答案。咖啡水应该_____。 Select the answer below that best completes the sentence. Water for coffee should _____. 没有氯/氯胺或铁 Have no chlorine/chloramine or iron 永远不要过滤或处理 Never be filtered or treated 包含高含量的铁 Contain high levels of iron 总是要软化到总硬度<30 ppm Always be softened to a total hardness < 30 ppm 问题编码 Question ID 100008282082 根据 SCA 水质手册，咖啡冲煮水的建议 pH 范围是多少？ According to the SCA Water Quality Handbook, what is the recommended pH range for coffee brewing water? 5 - 7 6.5 - 7 7 - 7.5 6 - 8 问题编码 Question ID 100008282083 某些地区的供水中钠过多的可能原因是什么？ What is the likely cause of excessive sodium in the water supply of some areas? 受到海水的污染 Seawater contamination of the water table 钙，镁和 pH Calcium, magnesium, and pH 碱度，总硬度和电导率 Alkalinity, total hardness, and electrical conductivity 钙，镁和碳酸盐 Calcium, magnesium, and carbonates	
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			produce acceptable results but may cause corrosion or limescale in brewing equipment. Note: Learners are often confused by the similarity between the terms alkaline, pH, and alkalinity. ASTs should anticipate questions in this area and be prepared to explain the difference.	问题编码 Question ID 100008282084 冲煮咖啡所用的三个主要的水质标度是什么？ What are the 3 central measures for water used for brewing coffee? 碱度，总硬度和 pH 值 Alkalinity, total hardness, and pH 钙，镁和 pH Calcium, magnesium, and pH 碱度，总硬度和导电率 Alkalinity, total hardness, and electrical conductivity 钙，镁和碳酸盐 Calcium, magnesium, and carbonates	
	3.03.07.03 碱度 Alkalinity	1. 使用 SCA Water Chart 定义咖啡冲煮的建议碱度范围 Define the recommended range of alkalinity for coffee brewing using the SCA Water Chart.	AST 应注意，SCA 水图使用行业标准单位 ppm CaCO_3 来量化总硬度和碱度。这不是错误。这可能会在一些学习者中引发问题，因此建议 AST 准备讨论。 ASTs should note that the SCA Water Chart uses industry-standard units, ppm CaCO_3, to quantify both total hardness and alkalinity. This is not an error. This may generate questions among some learners, so the AST is advised to be prepared to discuss.	问题编码 Question ID 100008282085 如果不考虑技术风险，为获得最佳冲煮水的建议碱度范围是多少？ What is the recommended range of water alkalinity to achieve optimal brews if technical risk is not a concern? 40-75 ppm CaCO_3 碱度 alkalinity 51-175 ppm CaCO_3 碱度 alkalinity 30-100 ppm CaCO_3 碱度 alkalinity 75-175 ppm CaCO_3 碱度 alkalinity	Wellinger & SCA, <i>The Water Chart</i>, 6 & 7 Wellinger & SCA, <i>SCA Water Quality Handbook</i>, 75

		2. 总结了冲煮水中的碱度与冲煮咖啡的预期感官之间的关系，以及在各种类型的冲煮设备上使用碱度过高或过低的水的可能影响。 Summarize the relationship between the level of alkalinity in brewing water and the expected sensory aspects of brewed coffee, as well as the possible effects of using water with overly high or low alkalinity on various types of brewing equipment.	问题编码 Question ID 100008282087 在什么情况下，碱度为 75 ppm CaCO_3 的水更有可能产生最佳的冲煮效果？ Under what circumstances, will water with an alkalinity of 75 ppm CaCO_3 more likely produce an optimal brew? <u>当水的总硬度在 150-175 ppm CaCO_3 的范围内时</u> When the total hardness of the water is in the range of 150 - 175 ppm CaCO_3. 当水的总硬度在 40-75 ppm CaCO_3 的范围 When the total hardness of the water is in the range of 40 - 75 ppm CaCO_3 。 当水的总硬度在 180-240 ppm CaCO_3 的范围 When the total hardness of the water is in the range of 180 - 240 ppm CaCO_3. 无论水的总硬度如何，都永远不会产生最佳的冲煮效果。 It never will produce an optimal brew regardless of total hardness of the water.	Wellinger & SCA, <i>The Water Chart</i>, 6 & 7 Wellinger & SCA, <i>SCA Water Quality Handbook</i>, 75 & Chapter 3
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				问题编码 Question ID 100008282089 碱度为 10 ppm CaCO_3 的水将如何影响饮料的味道，并对设备造成风险？ How will water with alkalinity of 10 ppm CaCO_3 likely affect the taste of the beverage, and would it pose risks to equipment? 饮料可能会变酸，并且有很高的腐蚀风险。The beverage will likely taste sour, and there is a high risk of corrosion. 该饮料可能会感觉到渣感和平乏，并且有较高的水垢风险。 The beverage will likely taste chalky and flat, and there is a higher risk of limescale. 饮料可能会感觉呈渣感和平乏，但没有技术风险。 The beverage will likely taste chalky and flat, but there is no technical risk. 饮料尝起来平衡，并且没有技术风险。 The beverage will likely taste balanced, and there is no technical risk.	
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				问题编码 Question ID 100008282090 碱度为 150 ppm CaCO_3 且总硬度为 250 ppm CaCO_3 的水将如何影响饮料的口味，并对设备构成风险？ How will water with alkalinity of 150 ppm CaCO_3 and total hardness of 250 ppm CaCO_3 likely affect the taste of the beverage, and would it pose risks to equipment? 饮料的味道可能是粉渣感且平乏的，并且形成水垢的风险较高; The beverage will likely taste chalky and flat, and there is a higher risk of limescale. 饮料的味道可能是酸味和尖锐的，而发生水垢的风险则较低; The beverage will likely taste sour and sharp, and there is a lower risk of limescale. 饮料的味道可能是变钝和酸的，并且存在较高的腐蚀的风险： The beverage will likely taste dull and sour, and there is a high risk of corrosion. 该饮料的口味可能均衡，没有技术风险。 The beverage will likely taste balanced, with no technical risk.	
	3.03.07.04 总硬度 (TH) Total Hardness (TH)	1. 使用 SCA Water Chart 定义咖啡冲煮的建议总硬度范围 Define the recommended range of total hardness for coffee brewing using the SCA Water Chart	AST 应注意，SCA 水图使用行业标准单位 ppm CaCO_3 来量化总硬度和碱度。这不是错误。这可能会在一些学习者中引发问题，因此建议	问题编码 Question ID 100008282092 在不存在腐蚀风险且避免结垢的情况下，当碱度为 40-65 ppm CaCO_3 时，最佳硬度的建议总硬度范围是多少？ Without corrosion risk and avoiding the cause of scale, what is the recommended range of total hardness to make an optimal brew when alkalinity is 40 - 65 ppm CaCO_3? 20-65 ppm CaCO_3 总硬度 total hardness 51-110 ppm CaCO_3 总硬度 total hardness 30-100 ppm CaCO_3 总硬度 total hardness 75-175 ppm CaCO_3 总硬度 total hardness	Wellinger & SCA, <i>The Water Chart</i>, 6 & 7 Wellinger & SCA, <i>SCA Water Quality Handbook</i>, 75 & Chapter 3
		2. 总结了冲煮水中总硬度水平与冲煮咖啡的预期感官之间的关系，并讨论了使用总硬度过高或过低的	AST 准备讨论。 ASTs should note that the SCA Water Chart uses industry-standard units, ppm CaCO_3, to quantify both total hardness and alkalinity. This is not an error. This may		

		水对各种类型的冲煮设备可能产生的影响 Summarize the relationship between the level of total hardness in brewing water and the expected sensory aspects of brewed coffee, as well as discuss the possible effects of using water with overly high or low total hardness on various types of brewing equipment	generate questions among some learners, so the AST is advised to be prepared to discuss.	问题编码 Question ID 100008282094 总硬度为 220 ppm CaCO_3 的水将如何影响饮料的味道? How will water with a total hardness of 220 ppm CaCO_3 likely affect the taste of the beverage? <u>咖啡会尝起来很重 Coffee will taste heavy</u> 咖啡会尝起来很粉末感 Coffee will taste chalky 咖啡会尝起来很弱 Coffee will taste weak 咖啡会尝起来很平乏 Coffee will taste flat	
				问题编码 Question ID 100008282095 总硬度为 20 ppm CaCO_3 的水将如何影响饮料的味道? How will water with a total hardness of 20 ppm CaCO_3 likely affect the taste of the beverage? 咖啡会尝起来很重 Coffee will taste heavy 咖啡会尝起来很粉末感 Coffee will taste chalky <u>咖啡会尝起来很弱 Coffee will taste weak</u> 咖啡会尝起来很平乏 Coffee will taste flat	
				问题编码 Question ID 100008282097 选择以下最能使句子完整的答案。将氯化钙添加到低碱度的水中 (CaCO_3 碱度 < 40 ppm) 进行冲煮。 Select the answer below that best completes the sentence. Adding calcium chloride to low alkalinity water (<40 ppm CaCO_3 alkalinity) for brewing will _____. <u>增加总硬度并可能增加腐蚀的风险 Increase total hardness and may increase risk of corrosion</u> 增加碱度但降低总硬度并减少腐蚀的风险 Increase alkalinity but decrease total hardness and reduce the risk of corrosion 仅增加碱度 Only increase alkalinity 仅增加总硬度 Only increase total hardness	

				问题编码 Question ID 100009234162 根据 SCA 水质手册，冲煮用水的总硬度和碱度之间的建议关系是什么？ According to the SCA Water Quality Handbook, what is the recommended relationship between total hardness and alkalinity in water for brewing? 总硬度应始终小于碱度，但不超过 40%，以降低腐蚀风险。 Total hardness should always be less than alkalinity, but by no more than 40% to reduce corrosion risks. 总硬度应始终小于碱度，但不应超过 80%，以降低腐蚀风险。 Total hardness should always be less than alkalinity, but by no more than 80% to reduce corrosion risks. <u>总硬度应始终超过碱度，但不超过 80%，以减少</u> <u>腐蚀风险。 Total hardness should always exceed alkalinity, but by no more than 80% to reduce corrosion risks.</u> 总硬度应始终超过碱度，但不超过 40%，以降低腐蚀风险。 Total hardness should always exceed alkalinity, but by no more than 40% to reduce corrosion risks.	
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	3.03.07.05 电导率 Electrical Conductivity	1. 定义咖啡冲煮的建议电导率范围 Define the recommended range of electrical conductivity for coffee brewing	EC / TDS 仪表通常以三种不同方式之一显示结果： 1. 作为 EC，以 $\mu\text{S} / \text{cm}$ 为单位（在当前的《水质手册》建议中引用） 2. 根据 TDS 估算，转换系数为 0.7。 3. 根据 TDS 估算，转换系数为 0.5 AST 应了解其仪表使用哪种测量方法，并准备根据需要应用适当的转换，以与电流电导率建议进行比较。	问题编码 Question ID 100008282098 选择以下最能使句子完整的答案。冲煮咖啡时，如果供水存在盐水污染的风险，建议的电导率范围（以 $\mu\text{S} / \text{cm}$ 为单位）应不超过_____。 Select the answer below that best completes the sentence. When brewing coffee, if the water supply has a risk of saltwater contamination, the recommended range of electrical conductivity (measured in $\mu\text{S}/\text{cm}$) should be no more than _____. 3 倍总碱度（以 ppm CaCO_3 为单位） 3 times alkalinity in ppm CaCO_3 3 倍总硬度（以 ppm CaCO_3 为单位） 3 times total hardness in ppm CaCO_3 0.7 碱度（以 ppm CaCO_3 为单位） 0.7 times alkalinity in ppm CaCO_3 0.5 倍总硬度（以 ppm CaCO_3 为单位） 0.5 times total hardness in ppm CaCO_3	Wellinger & SCA, <i>SCA Water Quality Handbook</i>, 22
			EC/TDS meters commonly display results in one of three different ways: 1. As EC, in units of $\mu\text{S}/\text{cm}$ (referenced in current Water Quality Handbook recommendations) 2. As estimated TDS with a conversion factor of 0.7 3. As estimated TDS with a conversion factor of 0.5 ASTs should be aware of which method their meter uses and be prepared to	问题编码 Question ID 100008282099 在供水中有盐水污染风险的地区，建议对冲煮水中的盐进行测试的建议方法是什么？ In areas where there is risk of saltwater contamination of the water supply, what is the recommended method of testing for salt in water for brewing? 比较水的电导率与其碱度 Comparing the electrical conductivity of the water against its alkalinity 比较水的电导率与其总硬度 Comparing the electrical conductivity of the water against its total hardness 比较水的总硬度与其碱度 Comparing the total hardness of the water against its alkalinity 比较水的电导率与其 pH 值 Comparing the electrical conductivity of the water against its pH	

		2. 请解释电导率/TDS 在水测试中的优点与局限。Explain the benefits and limitations of EC/TDS measurement for water analysis 3. 识别不合适的电导率/TDS 测试状况。Identify situations where EC/TDS measurement is and is not appropriate	apply appropriate conversion as needed to compare with current conductivity recommendations.	问题编码 Question ID 100008282100 手持式 TDS 测量仪显示水质的哪一方面? Which aspect of water quality does a handheld TDS meter display? 碱度 (ppm) Alkalinity in ppm 所有溶解固体的真实总量 The true total amount of all dissolved solids <u>根据电导率估算的总溶解固体 An estimate total dissolved solids, based on the electrical conductivity</u> 总硬度加碱度 Total hardness plus alkalinity	Wellinger & SCA, <i>SCA Water Quality Handbook</i>, Chapter 2.5 Chapter 10 Chapter 11.2
				问题编码 Question ID 100008282101 根据《SCA 水质手册》，手持式电导率或 TDS 仪表可用于执行以下两 (2) 项任务中的哪一项? According to the SCA Water Quality Handbook, a handheld electrical conductivity or TDS meter is useful for conducting which of the two (2) tasks listed below? <u>监测未经处理的水的稳定性 Monitoring the stability of untreated water</u> <u>监测经过处理的水的稳定性 Monitoring the stability of water after being treated</u> 将样品的总溶解固体作为独立的质量参数进行测量 Measuring the total dissolved solids of a sample as a standalone quality parameter 测量水样品的总硬度 Measuring the total hardness of a water sample	

	3.03.07.06 调整方法 Treatment Methods	1. 比较并对比以下旨在降低冲煮水中总硬度，碱度或其他物质的处理选项： - 反渗透（RO） - 泥沙过滤器 - 碳过滤器 - 柔软剂 - 脱碳机 - 脱碱剂 Compare and contrast the following treatment options aimed at reducing the total hardness, alkalinity, or other substances in water for brewing: - Reverse Osmosis (RO) - Sediment filter - Carbon filter - Softener - Decarboniser - De-alkalizer	详细的水处理系统设计超出了本课程的范围。专业水平的学习者只需了解水处理系统功能的主要样式以及对水化学的预期影响。这使学习者可以针对给定的情况建议一种系统。 掺入不合适的水的饮料可能会导致设备损坏。 建议 AST 提醒学习者参考咖啡设备制造商的水规格。具体的水处理系统设计问题应咨询合格的水系统技术人员。 Detailed water treatment system design is beyond the scope of this curriculum. The learner at professional level need only be aware of the main styles of water treatment systems function and expected effect on the water chemistry. This enables the learner to recommend a type of system for a given situation. Brewing with improper water compositions can result in equipment damage. ASTs are advised to remind learners to refer to coffee equipment manufacturer's water specifications. Specific water	问题编码 Question ID 100008282102 如果水中有微量氯，总硬度为 75ppm，碱度为 50ppm，建议采取什么处理方法？ If water has chlorine traces and a total hardness of 75ppm and an alkalinity of 50ppm, what treatment is recommended? <u>碳过滤器，无其他过滤器 Carbon filter with no other filtration</u> 反渗透，带 10%旁路 Reverse osmosis with 10% bypass 碳过滤器加上钠软化 Sodium softener with carbon filter 剂脱碳 Decarbonization 问题编码 Question ID 100008282103 如何在反渗透水处理中使用旁路？ How is bypass used in reverse osmosis water treatment? 防止所有未经处理的水进入反渗透系统。 All untreated water is prevented from entering the RO system. 处理过的水以相反的方向通过反渗透膜。 Treated water passes through the RO membrane in reverse direction. <u>预反渗透水用于将反渗透水重新矿化成所需的 TDS。 Pre-RO water is used to re-mineralize the RO water to desired TDS.</u> 一部分水绕过膜并用于冲洗系统中的杂质。 A percentage of water bypasses the membrane and is used to flush impurities from the system.	Wellinger & SCA, <i>SCA Water Quality Handbook</i>, Chapter 4
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			treatment system design questions should be referred to qualified water systems technicians.	问题编码 Question ID 100008282104 所有水都需要软化。选择对此语句的正确答案。 All water needs to be softened. Choose the correct response to this statement. 是的，硬度是设备和咖啡萃取的问题。YES, hardness is a problem for equipment and coffee extraction. 否，首先测量水并评估是否需要水处理。NO, first measure the water and evaluate if any water treatment is required. 是的，总硬度和碳酸盐（缓冲液）应始终降至最低。YES, total hardness and carbonates (buffer) should always be reduced to a minimum. 是的，设备供应商不会接受设备上没有软化过滤器的任何保修索赔。YES, equipment suppliers will not accept any warranty claims without a softener filter on the equipment.	
		2. 描述如何将两种或多种水混合以达到目标水组成 Describe how two or more waters can be blended to achieve target water compositions	分批混合水可以是生产冲煮水的有用技术，例如蒸馏水。将去离子水与过高的 TH 和碱水混合。此实践也可用于演示处理系统旁路功能的工作原理 Batch blending waters can be a useful technique to create brewing water, e.g. combining DI water with excessively high TH and alkalinity water. This practice can also be used to demonstrate how a treatment system bypass feature works.	授课，不考。Taught, not tested.	

	3.03.07.07 水质测试 Water Testing	1.确定并正确使用建议的工具来测试水的碱度，总硬度，pH 和潜在的钠过量 2.在 SCA Water Chart3 上绘制碱度和总硬度的样品值 3.讨论值是否在建议范围内 4.分析水以确定冲煮的适用性（根据建议），然后预测对冲煮咖啡的感官属性和冲煮设备健康的影响 1. Identify, and correctly use, the recommended tools to test water for alkalinity, total hardness, pH, and potential excess of sodium 2. Plot sample values for alkalinity and total hardness on the SCA Water Chart 3. Discuss whether values are within recommended ranges 4. Analyze water to determine suitability for brewing (according to recommendations), and then predict the effect on brewed coffee's sensory attributes, and brewing equipment health	可以使用滴数滴定套件（建议）或变色条来测量总硬度。请注意，一些测量套件制造商将其测试套件作为钙硬度测试来销售，尽管这些套件也可测量钙和镁的含量。 可使用滴数滴定套件（建议）或变色条来测量碱度。请注意，一些测量套件制造商将其测试套件作为碳酸盐硬度测试来销售，尽管这些套件也可以测量碱度。 测试套件可能无法以“ppm CaCO₃”的形式进行测量。绘图前，学习者必须使用 SCA 水质手册中所述的计算来转换其测量结果。 本活动中要测量的样品水应符合冲煮活动 3.05.03 中详述的规格。 Total hardness can be measured using drop count titration kits (recommended) or color-changing strips. Note that some measurement kit manufacturers market their test kits as calcium hardness	该主题有一个活动。 This topic has an activity. 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide	
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			tests, though these kits measure quantities of both calcium and magnesium. Alkalinity can be measured using drop count titration kits (recommended) or color-changing strips. Note that some measurement kit manufacturers market their test kits as carbonate hardness tests, though these kits do measure alkalinity. Test kits may not produce measurements in "ppm CaCO₃." Learners must convert their measurements using the calculation described in the SCA Water Quality Handbook before plotting. Sample waters to be measured in this activity should meet the specifications detailed in brewing activity 3.05.03.		
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	3.03.07.08 验证感官结果 Verifying Sensory Outcomes	1. 使用总硬度和碱度低于和高于 SCA 建议范围的水冲煮咖啡 Brew coffee using waters with levels of total hardness and alkalinity below and above SCA recommended ranges 2. 对每种冲煮完成感官分析并比较结果 Complete a sensory analysis of each brew and compare the results 3. 制定冲煮计划，以改善在 SCA 建议的总硬度和碱度水平下用水冲煮的咖啡的感官特性 Formulate brew plans which would improve the sensory characteristics of coffee brewed with water outside SCA recommended levels of total hardness and alkalinity 4. 测试这些修改后的冲煮计划中的至少一个，完成所得冲煮的感官分析，并与原始冲煮计划的结果进行比较 Test at least one of these modified brew plans, complete a sensory analysis of the resulting brew, and compare with the results from the original brew plan	该主题有一个活动 This topic has an activity 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide	
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3.03.08 Filter Media		1.认识到各种过滤介质之间的关系，以及它们冲煮之间的感官特性 Recognize the relationship between various filtering media, and the sensory characteristics between their brews 2. 意识到当使用相同的冲煮设备但使用不同的过滤介质时，通过咖啡床的流速可能会受到影响（例如，使用滴滤器或真空设备时） Aware that when using the same brewing device, but different filtering media, the flow rate through the coffee bed may be affected (e.g. when using drip filter or vacuum devices)		问题编码 Question ID 100008282105 在所有其他参数都相同的情况下，如果使用金属筛（網）代替滤纸，则完成的冲煮的哪些特性会有所不同？ With all other parameters being equal, if using a metal sieve (filter) instead of a paper filter, what characteristics of the finished brew would be different? 较低的温度 Lower temperature 口感增加 Increased body 甜度增加 Increased sweetness 更加干净 More clarification 问题编码 Question ID 100008282106 如果使用滤纸冲煮咖啡，则可能发生以下哪项情况？ If coffee is made using a paper filter, which of the following can be expected to occur? 由于液体较清澈，口感也较乾淨。 A clean taste due to the clarification of the beverage. 由于纸张中存在的氯而导致的颜色较浅， A lighter color due to chlorine present in paper. 由于成品中存在着芳香咖啡油，因此风味较强烈。 An intense taste due to the presence of aromatic coffee oils in the beverage. 以上所有 All of the above	SCAA & Lingle, <i>The Coffee Brewing Handbook</i>, 41-42 Rao, <i>Everything but Espresso</i>, 11
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				问题编码 Question ID 100008282107 哪种过滤介质最有可能使咖啡冲煮出最高醇厚度，最不透明的风味？ Which of these filtering media will most likely result in a brew with the most body and least flavor clarity? 金属 Metal 滤纸 Paper 滤布 Cloth 尼龙 Nylon	
				问题编码 Question ID 100008282108 清洁和存放滤布的最重要考虑因素是什么？ What is the most important consideration for the cleaning and storage of cloth filters? 防止腐坏的油和霉菌污染冲煮 Preventing rancid oils and molds from contaminating your brews 保留布料以延长其使用寿命 Preserving the cloth to extend its usable life 使用无洗涤剂的水 Use of water without detergents 快速清洁和快速干燥 Fast cleaning and quick drying	
			实践考试中考核。 Tested on Practical Exam. 该主题有一个活动。 This topic has an activity. 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide		

3.04 |栏目|冲煮过程 BREWING PROCESS

1 主题

主题 Topic	子主题 Sub-Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考文献 Reference
3.04.01 冲煮设备的使 用—重力冲煮器 Brewing Device Usage – Gravity Brewers	该主题是在中级水平上教授和测试的。 This topic was taught and tested at Intermediate level.				
3.04.02 冲煮工艺 Brewing Processes	该主题是在中级水平上教授和测试的。 该主题是在中级水平上教授和测试的 This topic was taught and tested at Intermediate level.				
3.04.03 润湿 (闷蒸) Wetting (Blooming)	3.04.03.01 水量 Water Quantity	1. 确定实现彻底润湿所需的最少水量 Identify the minimum amount of water required to achieve thorough wetting	学习者应该已经意识到, 在萃取阶段之前必须进行完全润湿, 润湿阶段所用的水量 (预润湿百分比), 闷蒸的持续时间以及在此期间施加的扰流的类型和润湿的水量会影响成品冲煮。	问题编码 Question ID 100008282109 通常来说, 每冲煮一克的咖啡大约需要多少水来润湿它? As a general rule, it typically requires a minimum of how much water for every gram of coffee to achieve thorough wetting. 2 克 2 grams 2.5 克 2.5 grams 1 克 1 gram 3 克 3 grams	SCAA & Lingle, <i>The Coffee Brewing Handbook</i> , 34 SCAA & Emma Sage, <i>Coffee Brewing: Wetting, Hydrolysis, Extraction Revisited</i>

			咖啡颗粒接受水润湿的能力将受到以下因素的影响： -咖啡产地 -烘焙程度和方法 -生豆和熟豆的新旧程度 Learners should already be aware that complete wetting must occur before the extraction phase can progress. The amount of water used for the wetting phase (pre-wet %), the duration of the pause for blooming, and the type and amount of turbulence applied during the wetting will affect the finished brew. A coffee particle's ability to accept water for wetting will be affected by factors such as: - Coffee origin - Roast level & methodology - Age of green and roasted coffee.	问题编码 Question ID 100008282110 使用滴滤过滤器冲煮机，达到完全润湿 20 克咖啡所需的最低水量是多少？ Using a drip filter brewer, what is the minimum amount of water required to achieve a thorough wetting of a 20 gram coffee dose? 40 克 40 grams 45 克 45 grams 20 克 20 gram 60 克 60 grams	
	3.04.03.02 吸水性 Wettability	1. 意识到不同的润湿阶段方法与成品冲煮之间的关系 Aware of factors that influence a coffee particle's ability to accept water for wetting		授课，不考。 Taught, not tested.	
	3.04.03.03 基于保水率计算萃取率 Calculating Yield Based on Water Retention	1. 根据萃取咖啡床的典型液体保留量计算饮料萃取率 Calculate beverage yield based on typical liquid retention of the extracted coffee bed		问题编码 Question ID 100008282114 如果用 2 升水和 1:17 的冲煮比例冲煮咖啡，则预计冲煮的最终量约为多少？ If a coffee is brewed with 2 liters of water and a 1:17 brew ratio, what is the approximate expected volume of finished brew? 2.5 升 2.5 liters 1.5 升 1.5 liters 1.8 升 1.8 liters 1.6 升 1.6 liters	

			问题编码 Question ID 100008282115 如果用 1.2 升水和 70 克磨成粉的咖啡冲煮咖啡，预计饮料的大概萃取量是多少？ If a coffee is brewed with 1.2 liters of water and 70 grams of ground coffee, what is the approximate expected beverage yield? 1.2 升 1.2 liters 1.13 升 1.13 liters 1.06 升 1.06 liters 1.0 升 1.0 liters	
	3.04.03.04 润湿阶段与成品 冲煮的关系 Wetting Phase and Finished Brew Relationship	1. 意识到不同的润湿阶段方法与成品冲煮之间的关系 Aware of the relationship between different wetting phase methodologies and differences in the finished brew	问题编码 Question ID 0000000008282116 润湿不均匀会导致以下哪些情况发生？ Non-uniform wetting will cause which of the following to occur? 萃取不足 Under extraction 萃取过度 Over extraction 萃取不一致 Inconsistent extraction 不会影响萃取 No influence on extraction	

			问题编码 Question ID 100008282117 使用滴濾式沖煮方式，將 60 克的水添加到 15 克的咖啡粉中，短暫暫停進行悶蒸之後，再注入剩餘的水完成沖煮，可能的結果是？ Using a drip filter brewer, 60 grams of water is added to a ground coffee dose of 15 grams. Before adding the remaining water to complete the brew, a pause is taken to allow for the bloom. What is the likely result? <u>咖啡粉床太湿了，过量的水通过咖啡粉床，会导致稀释成品浓度，并可能降低萃取率。 The coffee bed is overly wet. Excess water is likely to pass through the bed, diluting the brew, and reducing potential solubles yield.</u> 实现了最佳的润湿效果，并可能产生均匀的萃取。 Optimal wetting is achieved and will likely produce an even extraction. 可接受的咖啡润湿度已达成，但可能会产生风味较强的冲煮成品。 Acceptable wetting of the coffee is achieved, but will likely produce a stronger brew. 可接受的咖啡润湿度达成，但可能会延长接触时间。 Acceptable wetting of the coffee is achieved , but will likely extend the contact time.	
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			问题编码 Question ID 100008282118 使用滴濾式沖煮方式，將 30 克的水添加到 20 克的咖啡粉中，短暫暫停進行悶蒸之後，再注入剩餘的水完成沖煮，可能的結果是？ Using a drip filter brewer, 30 grams of water is added to a ground coffee dose of 20 grams. Before the remaining water is added to complete the brew, a pause for the bloom is allowed. What is the likely result? 咖啡粉并没有彻底的被润湿，可能会导致不均匀的萃取。 <u>It is unlikely that a thorough wetting will be achieved, and it is likely that an uneven extraction will be produced.</u> 很可能会得到充分的润湿和均匀的萃取。 It is likely that a thorough wetting and an even extraction will be produced. 咖啡粉会过度湿润，萃取阶段提早开始，咖啡很可能会过度萃取。 The coffee will be overly wet, beginning the extraction phase sooner, and the coffee is likely to be over extracted. 咖啡粉会过度湿润，咖啡萃取阶段延迟，导致咖啡萃取不足。 The coffee will be overly wet, delaying the extraction phase, and the coffee will be under extracted.	
	3.04.03.05 润湿应用 Wetting Application	1. 确定应用润湿阶段的不同方法 Identify different methods of applying the wetting phase	该主题有一个活动。 This topic has an activity. 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide	

3.05 |栏目|冲煮分析| BREWING ANALYSIS

2 主题

主题 Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考文献 Reference
3.05.01 描述冲煮 Describe the Brew	该主题是在中级水平上教授和测试的 This topic was taught and tested at Intermediate level.			
3.05.02 平衡冲煮 Balanced Brew	该主题是在中级水平上教授和测试的 This topic was taught and tested at Intermediate level.			
3.05.03 溶出效率 (萃取) Solubles Yield (Extraction)	1. 制定策略以改变咖啡与水的比例以及其他必要的冲煮元素，以增加或减少萃取量而不会改变强度 2. 使用此策略以不同的萃取量和相同的强度冲煮咖啡 3. 总结冲煮咖啡的感官方面与萃取水平之间的关系 1. Develop a strategy for changing coffee to water ratio and other essential elements of brewing to increase or decrease extraction without changing strength 2. Use this strategy to brew coffees at various extractions and the same strength 3. Summarize the relationships between the sensory aspects of the brewed coffee and level of extraction		问题编码 Question ID 100008282119 具有以酸度为主，缺少醇厚度的冲煮如何改善？ How can a brew that had dominant sour tastes and lacks body be improved? 增加萃取 Increase extraction 减少萃取 Decrease extraction 降低水温 Decrease water temperature 粗研磨 Grind coarser 问题编码 Question ID 100008282120 冲煮的咖啡有复杂的酸度、低甜和低苦，需要怎么调整才能让这个咖啡变更好喝？ How can a brew that had complex acidity, low sweetness, and very low bitterness be improved? 增加萃取，直到甜度最高，苦度可接受 Increase extraction until sweetness is maximized and bitterness is acceptable. 不太可能改进，不需要调整。 It is unlikely this brew could be improved. No adjustment is necessary. 降低萃取，增加酸度，排除苦味 Decrease extraction to increase acidity and eliminate bitterness. 提高浓度 Make the brew stronger.	SCAA & Lingle, <i>The Coffee Brewing Handbook</i> , 10 & 12

			问题编码 Question ID 100008282121 以下哪项可能降低可溶物的萃取率? Which of the following has the potential to reduce the solubles yield? 将水温从 94°C/201°F 更改为 96°C/205°F 更细的研磨设置 Finer grind setting 更多的接触时间 More contact time 将咖啡与水的比例从 1:17 更改为 1:14 <u>Changing the coffee to water ratio from 1:17 to 1:14</u>	
			问题编码 Question ID 100008282122 要显著改变萃取率百分比，但保持相同的强度，应更改以下哪项？ To significantly change the extraction %, but keep the same strength, which of the following should be changed? 温度和过滤设备研磨 Temperature and filtering device 研磨刻度和水粉比 Grind setting and coffee to water ratio 咖啡水比和扰流 Coffee to water ratio and turbulence 扰流和温度 Turbulence and temperature	

		此活动（请参阅下面的附录 8 –活动准备指南）旨在通过以下方式提高学习者优化冲煮咖啡风味的能力： 1. 使用现有的咖啡，磨豆机和冲煮设备提高其增加和减少萃取的能力 2. 允许学习者收集和解释不同萃取水平下有关咖啡的感官观察结果。 This activity (see Appendix 8 – Activities Preparation Guide below) is designed to increase the learner's ability to optimize the flavor of a brewed coffee by: 1. Improving their ability to increase and decrease extraction using the available coffees, grinders, and brewing devices 2. Allowing the learner to collect and interpret sensory observations about the coffee at different extraction levels.	在实践考试中考核。 Tested on Practical Exam. 该主题有一个活动。 This topic has an activity. 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8. Appendix - Activities Preparation Guide	
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3.05.04 可溶性物质浓度 (强度) Solubles Concentration (Strength)	1. 制定策略, 根据需要更改咖啡与水的比例以及其他必要的冲煮元素, 以增加或降低冲煮咖啡的强度, 而无需更改萃取 Develop a strategy for changing the coffee to water ratio and other essential elements of brewing as needed to increase or decrease brewed coffee strength without changing extraction		问题编码 Question ID 100008282123 如果以 1:16 的比例冲煮咖啡并测得浓度为 1.50%, 可以采取什么措施来降低浓度, 但要保持类似的萃取率? If a coffee is brewed at 1:16 and measured 1.50% strength, what could be done to reduce the strength, but keep a similar extraction? 改变水粉比- 增加粉量 Change the coffee to water ratio by increasing the dose. 改变水粉比- 减少粉量 Change the coffee to water ratio by decreasing the dose. 改变研磨设置- 增加接触时间 Change the grind setting to increase contact time. 改变研磨设置- 减少接触时间 Change the grind setting to decrease contact time. 问题编码 Question ID 100008282124 滴滤咖啡的浓度为 1.3%, 配方如下: A drip filter coffee has a 1.3% strength with the following recipe: -20 克磨碎的咖啡- 20 grams of ground coffee -320 克水- 320 grams of water 如果将研磨的咖啡剂量增加到 21.3 克, 并且未更改其他冲煮参数, 则大概的咖啡浓度是多少? If the ground coffee dose is increased to 21.3 grams, and no other brewing parameters are changed, what is the likely approximate strength of the brew? 1.4% 1.45% 1.3% 1.22%	SCAA & Lingle, <i>The Coffee Brewing Handbook</i>, 10 & 12
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		问题编码 Question ID 100008282125 滴滤咖啡按照以下配方进行调制: -20 克磨碎的咖啡 -320 克水 如果最终浓度为 1.3%，应进行哪些更改以将强度提高至 1.4%，并保持类似的可溶物萃取率？ A drip filter coffee is brewed with the following recipe: - 20 grams of ground coffee - 320 grams of water If the resulting strength is 1.3%, what change should be made to increase the strength to 1.4%, and maintain a similar solubles yield? 将磨碎的咖啡量增加至 20.6 克并进行细研磨;Increase ground coffee amount to 20.6 grams and grind finer. 将磨碎的咖啡增加至 21.3 克;Increase ground coffee amount to 21.3 grams. 将磨碎的咖啡增加至 22.8 克并进行粗研磨;Increase ground coffee amount to 22.8 grams and grind coarser. 将接触时间增加至 3:30 并进行细研磨。Increase contact time to 3:30 and grind finer.	
		问题编码 Question ID 100008282126 滴滤咖啡的配方浓度为 1.41%，其配方如下：A drip filter coffee achieves a 1.41% strength with the following recipe: -20 克咖啡粉 20 grams of ground coffee -300 克水 300 grams of water 如果将磨碎的咖啡剂量减少到 17.6 克且未更改其他冲煮参数，则该咖啡的大概强度是多少？ If the ground coffee dose is reduced to 17.6 grams and no other brewing parameters are changed, what is the likely approximate strength of the brew? 1.4% 1.35% 1.3% <u>1.22%</u>	

	2.使用此策略以各种强度和相同的萃取量冲煮咖啡 3。概括所用咖啡的感官与冲煮强度之间的关系 2. Use this strategy to brew coffees at various strengths and the same extraction 3. Generalize the relationships between the sensory aspects of the coffee used and brew strength	Tested on Practical Exam. 经过实践考试。 This topic has an activity. 该主题有一个活动。 See instructions in 8.Appendix - Activities Preparation Guide 请参阅第 8 节中的说明。附录-活动准备指南	
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3.06 | Section | MAINTENANCE

1 主题

Topic	子主题 Sub-Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考文献 Reference
3.06.01 磨豆机刀盘 Grinder Burrs	3.06.01.01 刀盘边缘质量 Burrs Edge Quality	1.意识到磨豆机刀盘边缘质量对颗粒质量和粒径分布的影响 Aware of the effect of the grinder burr edge quality on particle quality and size distribution 2.认识到磨豆机刀盘边缘质量和成品咖啡之间的关系 Recognize the relationship between grinder burr edge quality and the finished brew	This topic has an activity. 该主题有一个活动。 See instructions in 8.Appendix - Activities Preparation Guide 请参阅第 8 节中的说明。附录-活动准备指南		
	3.06.01.02 更换要求 Replacement Requirement	1.识别出表明需要更换磨豆机刀盘或刀盘不平行的症状 Identify symptoms that indicate the grinder burrs require replacement, or burrs are not parallel		问题编码 Question ID 100008282127 假设使用相同的冲煮参数，相同的咖啡豆，下列哪一项表示磨豆机刀盘还不需要更换？ Assuming use of similar recipes, with familiar coffees, which of the following would NOT indicate that the grinder burrs may require replacement? <u>可溶性物质萃取率普遍增加，加上更多的酸味体现。</u> <u>A general increase in solubles yield, combined with more sour tasting brews.</u> 相同萃取率的冲煮通常变得更苦涩。 Brews generally become more bitter at the same solubles yield. 刀盘在研磨过程中会产生更多的噪音。 The burrs produce more noise during grinding. 泥巴似的咖啡粉床 Muddier coffee beds	Rao, <i>Everything but Espresso</i>, 8 & 9 Rao S., Barista Guild, <i>On Extraction Measurement</i>

				问题编码 Question ID 100008282128 下列哪些症状可能表明磨豆机刀盘损坏或不平行？ Which of the following symptoms may indicate the grinder burrs are damaged or unparallel? <u>使用相同的烘焙批次和配方，可溶物萃取率不一致；</u> <u>Using the same roast batch and recipe, the solubles yield in the batch brews are inconsistent.</u> 使用相同的配方搭配不同的咖啡，可溶物萃取率不一致。 Using the same recipe with different coffees, the solubles yield in the batch brews are inconsistent. 当更改冲煮器中的冲煮量大小，可溶物萃取率发生变化。 The solubles yield changes when the batch size in the batch brewer is changed. 上述所有的 All of the above	
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3.07 |栏目| 旁路 BYPASS

3 主题

主题 Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考文献 Reference
3.07.01 使用旁路的好处 Benefits of Using Bypass	1. 确定使用旁路的好处 Identify benefits of using bypass	旁路可用于提高大容量冲煮机的效率。这是通过以下方法实现的：通过使用冲煮设备的最大容量来生产具有更高可溶物浓度的冲煮器，然后用热水稀释更浓的冲煮器。 旁路技术已成功用于冲煮比赛中，以限制萃取并减少苦味。AST 应该解释使用旁路如何影响溶出速率，以及如何使用它来限制萃取。 尽管《咖啡冲煮手册》建议最多使用 40% 的旁路水，但没有设定最大旁路百分比的规则，应鼓励学习者探索范围广泛的旁路百分比。 Bypass can be used to increase the efficiency of a batch brewer. This is achieved by producing brews with a higher solubles concentration, utilizing the maximum capacity of the brewing device, and then diluting the stronger brew with hot water. Bypass has been used successfully in brewing competitions to limit extraction and reduce bitterness. ASTs should explain how dissolution rates are affected by using bypass, and how this can be used to limit extraction.	问题编码 Question ID 100008282129 以下内容哪些是对的？ Which of the following is true? 如果用 5L/1.2gal 的批量冲煮机器可以制作 4.3L/1.1gal 咖啡的话，我们可以运用旁路法在同样的时间内制作 6L/1.6gal 的咖啡。 <u>Bypass can be used to produce 6L/1.6gal of brewed coffee in the same amount of time it would take to produce 4.3L/1.1gal of brewed coffee from a 5L/1.3gal batch brewer.</u> 当使用旁路生产 3 升的冲煮咖啡时，与不使用旁路的 3 升的冲煮咖啡相比，需要更多的研磨咖啡。 When using bypass to produce 3 liters of brewed coffee, it will require more ground coffee than 3 liters of brewed coffee without bypass 旁路可以用于减少生产相同体积的冲煮咖啡所需的研磨咖啡量。 Bypass can be used to reduce the amount of ground coffee required to produce the same volume of brewed coffee. 尽管旁通改善了批量冲煮的效率，但是通常冲泡出劣质的咖啡。 Although bypassing improves the efficiency of a batch brewer, it typically produces inferior brews.	SCAA, Best Practice, <i>Guidelines for Using By-Pass in the Drip Coffee Brewing Process</i> SCAA & Lingle, <i>The Coffee Brewing Handbook</i>, 35

		Although the Coffee Brewing Handbook recommends that up to 40% of the water bypasses the coffee bed, there are no set rules for maximum bypass %, and learners should be encouraged to explore a wide range of bypass %.	问题编码 Question ID 100008282130 以下哪个说法是正确的？ Which statement below is correct? <u>旁路可用于减少成品冲煮中过度萃取的特征。 Bypass can be used to reduce over extracted characteristics in the finished brew.</u> 旁路冲煮可能由于接触时间减少而具有萃取特征不足的情况。 Bypass brews may have under extracted characteristics due to reduced contact time. 使用浸泡式冲煮机，在所有参数相同的情况下，旁路冲煮应与非旁路冲煮相比具有相同的味道特征。 Using an immersion brewer, with all parameters being equal, a bypassed brew should achieve identical taste characteristics compared with a non-bypassed brew. 旁路冲煮通常味道更浓，更苦。 Bypass brews typically taste stronger and more bitter.	
			问题编码 Question ID 100008282131 以下内容哪些是对的？ Which of the following is true? <u>旁路可用于改善萃取率良好但浓度过高的饮料。 Bypass can be used to improve a well extracted, but overly strong beverage.</u> 旁路仅可与滴滤过滤器一起使用。 Bypass can only be used with drip filter brewers. 旁路仅可与滴滤器和浸泡式冲煮器一起使用。 Bypass can only be used with drip filter and immersion brewers. 旁路冲煮通常味道较弱且较酸。 Bypass brews typically taste weaker and more sour.	

3.07.02 计算，测量和绕过 图表 Calculate, Measure, and Chart Bypass	1. 计算旁路百分比，并测量并绘制旁路冲煮图表 Calculate bypass %, and measure and chart a bypassed brew	旁路百分比与使用的总水量直接相关， 学习者应该能够计算出达到目标冲煮所需的旁路百分比，因为这在某些分批冲煮器中为编程旁路百分比时是必不可少的技能。 计算添加到完成冲煮中的水量，以将其可溶物浓度降低至所需强度。 对批量冲煮器进行编程时，必须考虑水的密度 The bypass % relates directly to the total amount of water used. The learner should be able to calculate the bypass % required to achieve a target brew, as this is a necessary skill when programming bypass % in some batch brewers. The learner should be able to calculate how much water to add to a finished brew to reduce its soluble concentration to a desired strength. Note: When programming batch brewers, the density of water must be taken into consideration.	问题编码 Question ID 100008282132 使用 60 克咖啡和 15% 的旁路水，以 1:16 的水粉比冲煮咖啡，需要在咖啡床上放多少水？To make a brew with 1:16 coffee to water ratio using 60 grams of coffee and a 15% bypass, how much water will need to be placed into the coffee bed? 960 克 960 grams 816 克 816 grams 860 克 860 grams 1000 克 1000 grams 问题编码 Question ID 100008282133 计算，测量和绘制旁路冲煮的正确过程顺序是什么？ What is the correct order of process for calculating, measuring, and charting a bypass brew? a. 选择旁路百分比 Choose bypass % b. 计算旁路之前的冲煮比 Calculate ratio for pre-bypass brew c. 煮咖啡 Brew coffee d. 测量并绘制绕过冲煮的图表 Measure and chart pre-bypassed brew e. 添加旁路水 Add bypass water (正确的顺序是 a, b, c, d, e.) (Correct order is a,b,c,d,e.)	SCAA, Best Practice, Guidelines for Using By-Pass in the Drip Coffee Brewing Process
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		问题编码 Question ID 100008282136 使用 20 克咖啡和 320 克水的滴滤过滤器冲煮可达到 1.45% 的浓度和 21% 的萃取率， 现在应向冲煮器中添加多少水以将其浓度降至 1.35% 并保持 21% 的萃取率？ A drip filter brew, using 20 grams of coffee, and 320 grams of water, achieves 1.45% strength, and 21% extraction. How much water should now be added to the brew to reduce its strength to 1.35%, and maintain 21% extraction? 20 克 20 grams 10 克 10 grams 6 克 6 grams 5 克 5 grams	
3.07.03 旁路对冲煮咖啡感官方面的影响 Effect of Bypass on the Sensory Aspects of Brewed Coffee	1. 意识到旁路和非旁路冲煮之间的感官差异，可达到相似的可溶物萃取率和可溶物浓度 Aware of sensory differences between bypass and non-bypass brews that achieve similar solubles yields and solubles concentration	This topic has an activity. 该主题有一个活动。 请参阅第 8 节中的说明。附录-活动准备指南 See instructions in 8.Appendix - Activities Preparation Guide	

5. 关键的 SCA 培训文件 Essential SCA Training Documents

- Is Your Water Suitable for Brewing
- Ratio Control Chart 101017
- Professional Practical Exam Worksheet
- Brew Recipe Modification Worksheet

所有文档可以在 AST 门户的课程与考试栏目中找到 All documents are available at the AST Portal under Curriculum and Exams/Brewing.

6. 所需设备和物料清单 Required Equipment and Supplies List

在 AST 门户中的资源板块 Available at the AST Portal under Resources/Venue Requirements.

SCA 英国或者美国商店中可以找到的东西均已提供链接 Any items available in the SCA US or UK store are noted and a link directly to the store is provided.

7. 参考书目 Bibliography

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8. 附录-活动准备指南 Appendix - Activities Preparation Guide

- **活动和实践考试重要注意事项：**

- AST **必须**为实践考试提供与活动 3.03.07.08 中使用的咖啡不同的咖啡。
- 在实践考试和活动 3.05.03 和 3.05.04 **中使用的咖啡必须**能够产生 24%的可溶物产率（萃取率）。

- **Important Note for Activities and Practical Exam:**

- The AST **must** provide a coffee for the practical exam that is different from the coffee used in Activity 3.03.07.08.
- The coffee used in the practical exam and activities 3.05.03 & 3.05.04 **must** be capable of producing a solubles yield of 24%.

编码 Code	说明 Instructions
3.01.02 烘焙程度 Roast Level	选择一种咖啡，烘焙到 3 种不同的水平（每杯咖啡之间至少有 3% 的差异）： ● 咖啡 1-Agtron“商用”45 至 50，发展 13%至 17% ● 咖啡 2-Agtron“商用” 50 至 55，发展 9%至 13% ● 咖啡 3-Agtron“商用” 40 至 45，发展 17%至 21% 注意：烘焙颜色应使用符合 SCA 杯装标准的咖啡粉进行测量。 Select one coffee, roasted to 3 different levels (with at least 3% development difference between each coffee): ● Coffee 1 - Agtron 'Commercial' 45 to 50, with 13% to 17% development ● Coffee 2 - Agtron 'Commercial' 50 to 55, with 9% to 13% development ● Coffee 3 - Agtron 'Commercial' 40 to 45, with 17% to 21% development Note: Roast color is to be measured with coffee ground to the SCA Cupping Standard. 学习者将通过浸泡式和滴滤式冲煮设备，品尝同一种烘焙到 3 种不同水平的咖啡，并进行以下比较： ● 所有冲煮之间的感官特征 ● 浸泡冲煮之间的可溶性溶解物浓度 ● 滴滤冲煮之间的可溶性物质浓度和萃取率 ● 滴滤冲煮之间的总接触时间 Learners will taste the same coffee roasted to 3 different levels, from an immersion and a drip filter device, and compare the following: ● Sensory characteristics between all brews ● Solubles concentration between immersion brews ● Solubles concentration and solubles yield between drip filter brews ● Total contact time between drip filter brews Learners will then prepare a plan to improve the brews of Coffees 2 and 3, using either the drip filter brewer, or the immersion brewer.

第 1 部分

- 使用浸泡式设备制作参照冲煮咖啡 1，其配方可以使平衡度，风味和强度达到令人满意的平衡。
- 测量可溶物浓度，然后品尝并描述咖啡 1。
- 使用相同的配方以冲煮咖啡 2 和咖啡 3。
- 测量来自咖啡 2 和咖啡 3 中的可溶物浓度。
- 品尝和描述咖啡 2 和咖啡 3。
- 讨论咖啡 1，咖啡 2 和咖啡 3 之间的所有差异。

Part 1

- Use an immersion device to produce a reference brew of Coffee 1, using a recipe that produces a pleasant balance of taste, flavor, and strength.
- Measure solubles concentration, then taste and describe the brew from Coffee 1.
- Use the same recipe to produce a brew with Coffee 2, and a brew with Coffee 3.
- Measure solubles concentration of brews from Coffee 2 and Coffee 3.
- Taste and describe brews from Coffee 2, and Coffee 3.
- Discuss all differences between brews from Coffees 1, 2, and 3.

第 2 部分

- 使用滴滤器设备制作参照咖啡，其配方采用能够使平衡度，风味和强度达到令人愉悦的平衡的配方。记录总的接触时间至关重要。
- 测量可溶物浓度，然后品尝并描述咖啡 1。
- 使用相同的配方以咖啡 2 冲煮和咖啡 3 冲煮。
- 记录闷蒸的任何差异，并记录来自咖啡 2 和咖啡 3 的冲煮咖啡的总接触时间。
- 测量来自咖啡 2 和咖啡 3 的冲煮物中的可溶物浓度。
- 品尝和描述咖啡 2 和咖啡 3。
- 讨论咖啡 1，咖啡 2 和咖啡 3 的冲煮之间的所有差异。

Part 2

- Use a drip filter device to produce a reference brew of Coffee 1, using a recipe that produces a pleasant balance of taste, flavor, and strength. It is critical that the total contact time is recorded.
- Measure solubles concentration, then taste and describe the brew from Coffee 1.
- Use the same recipe to produce a brew with Coffee 2, and a brew with Coffee 3.
- Note any differences in the bloom and record the total contact times of the brews from Coffee 2 and Coffee 3.
- Measure solubles concentration of brews from Coffee 2 and Coffee 3.
- Taste and describe brews from Coffee 2, and Coffee 3.
- Discuss all differences between brews from Coffees 1, 2, and 3.

第三部分

- 学习者选择浸泡式或滴滤式冲煮器，并制定计划以改进咖啡 2 和咖啡 3 的冲煮方式。
- 然后，学习者使用这些计划准备至少再冲煮 2 号咖啡和 3 号咖啡。

	• 测量冲煮并将感官特性与第 1 部分或第 2 部分中的冲煮进行比较。 • 讨论结果。 Part 3 • Learners select either the immersion or the drip filter brewer and create plans that improve the brews for Coffee 2 and Coffee 3. • Learners then use these plans to prepare at least one additional brew of Coffee 2, and Coffee 3. • Measure the brews and compare the sensory characteristics against the brews in Part 1 or Part 2. • Discuss the results.
3.02.02.03 设备形状对冲煮成品的影响 Effect of Device Shape on the Finished Brew	学习者将使用由相同材料（例如陶瓷）制成但形状不同（例如圆锥形，截头圆锥形，平底形）的设备，评估采用相同配方制备的冲煮冲煮之间的感官差异。 Learners will evaluate the sensory differences between brews prepared with an identical recipe, using devices made of the same material (e.g. ceramic), but of different shapes (i.e. cone, truncated cone, flat bottom). 第 1 部分 • 学习者可以使用锥形滴滤器设备准备参考容量为 400 毫升的参照冲煮，该设备可实现宜人的咖啡味道并且萃取率得当，最低可溶物浓度为 1.15%，萃取后完成对饮料的感官评估。 • 学习者应用相同的配方，分别使用锥形冲煮设备和平底冲煮设备中制作冲煮，然后完成对这些冲煮的感官评估。 注意：对于所有 3 种设备使用相同的配方时，学习者必须应用相同的技术，这一点至关重要，尤其是以下方面： • 预浸泡时间和水 % • 注水时间 • 注水高度 • 水温 第 2 部分 根据以上比较，学习者需要制定计划，以改善其中 2 个冲煮效果较差的咖啡的结果。 Part 1 • Learners prepare a reference brew of up to 400 ml using a cone drip filter device, that achieves a pleasant tasting solubles yield, with a minimum solubles concentration of 1.15%, and then complete a sensory evaluation of the brew. • Learners then apply the same recipe to create brews in the truncated cone, and flat bottom brewers, and then complete sensory evaluations of those brews. Note: It is critical that learners apply identical techniques when using the same recipe for all 3 devices, especially the following aspects: • Pre-wet duration and water % • Water delivery time • Pouring height • Water temperature Learners compare the sensory evaluations of the brews from each device and discuss how the recipe for each device could be modified to improve the results. Part 2 Based on the comparison above, the learners then create plans to improve the results from 2 of the brewers that produced less favorable results.

3.03.03.04
Effect of Particle
Size Distribution
on the Brew

研磨咖啡的样品可以在使用前不久制备，并在必要时存储在密封的容器中。

用于本练习的所有样品（包括参照/校准样品）必须同时准备。

以浸泡式冲煮器的参照/校准研磨设置研磨咖啡。将此样品用作参照冲煮。

使用相同的研磨设置研磨额外的咖啡，然后使用 800 微米和 400 微米筛子将咖啡粉分成三个不同的大小范围：

- 400 微米以下
- 在 400 至 800 微米之间
- 800 微米以上

学习者使用浸泡式冲煮设备比较从不同粒径分布范围制备的冲煮咖啡。

学习者制定冲煮计划，并准备以下冲煮方法：

- 冲煮 1-参照冲煮
- 冲煮 2-咖啡粉低于 400 微米
- 冲煮 3-咖啡粉在 400 至 800 微米之间
- 冲煮 4-咖啡粉超过 800 微米

AST 应为此练习选择可以连续使用的冲煮设备，并保持其他参数（包括接触时间）。

每种冲煮物中的可溶物浓度应使用折光仪测量，并比较结果。

- 学习者应使用 Coffee Taster 风味轮中的描述符记录每种咖啡的感官观察结果（包括香气，味道，口感和颜色）。
- 讨论结果

Samples of ground coffee can be prepared shortly before the course and stored in an airtight container if necessary.

All samples used for this exercise (including the reference/calibration sample), must be prepared at the same time.

Grind coffee at a reference/calibration grind setting for the immersion brewer. Use this sample for your reference brew.

Grind additional coffee using the same grind setting, and then use 800 micron and 400 micron sieves to separate the grounds into three different ranges of sizes:

- Below 400 microns
- Between 400 and 800 microns
- Above 800 microns.

Learners use an immersion brewing device to compare brewed coffees prepared from different particle size distribution ranges.

Learners formulate brew plans, and prepare the following brews:

- Brew 1 - Reference brew
- Brew 2 - Coffee grounds below 400 microns
- Brew 3 - Coffee grounds between 400 and 800 microns
- Brew 4 - Coffee grounds above 800 microns

The AST should select a brewing device for this exercise that can be used consistently, and maintain other parameters including contact time.

The solubles concentration of each brew should be measured with a refractometer, and the results compared.

- Learners should record sensory observations (including aroma, taste, body, and color) for each coffee, using descriptors from the Coffee Taster's Flavor Wheel.

	Discuss the results.
3.03.04 冲煮时间 Brewing Time	学习者使用浸泡式冲煮器制备参照冲煮，其最低可溶物浓度为 1.15%，建议的可溶物萃取率为 18%至 22%。 - 冲煮 1 = 1 分钟的接触时间 - 冲煮 2 =接触时间比参照冲煮时间短 1 分钟 - 冲煮 3 =接触时间比参照冲煮多 1 分钟 学习者测量 TDS，并完成每种冲煮的感官分析。比较并讨论所有 4 种冲煮之间的 TDS 和感官差异。 Learners prepare a reference brew with a minimum solubles concentration of 1.15%, and a recommended solubles yield of 18% to 22%, using an immersion brewer. Learners complete a sensory analysis of the reference brew, and then prepare 3 additional brews with identical parameters, except contact time as follows: - Brew 1 = 1 minute contact time - Brew 2 = Contact time of 1 minute less than reference brew - Brew 3 = Contact time of 1 minute more than reference brew Learners measure the TDS, and complete sensory analysis of each brew. Compare and discuss TDS and sensory differences between all 4 brews.
3.03.05.02 水温-对煮咖啡感官方面的影响 Water Temperature - Effect on Sensory Aspects of Brewed Coffee	学习者制定冲煮计划，并使用冲煮设备以 91°C，93°C和 96°C的水冲煮咖啡，而所有其他冲煮参数均相同。对于每种咖啡，学习者应记录感官观察结果，使用折光仪进行测量，并使用咖啡冲煮控制图进行分析。请注意，在更高的海拔高度可能无法达到 96°C的水温。AST 可以根据需要调整此练习。 Learners formulate brew plans and use a brewing device to brew coffees using water with temperatures of 91°C, 93°C, and 96°C with all other brew parameters the same. For each coffee, learners should record sensory observations, measure using a refractometer, and analyze using the coffee brewing control chart. Note that achieving 96°C water temperature may not be possible at higher elevations. ASTs can adapt this exercise as needed.
3.03.06 冲煮扰流 Brew Turbulence	第 1 部分：浸泡扰流比较 - 学习者准备和分析（根据 TDS 和感官观察结果）“参照”浸泡冲煮，该冲煮使用 1：14.5 的比例并达到 1.30%（+/- 0.05%）的可溶性浓度。在准备参照冲煮时，学习者可以施加任何类型的合适扰流。 - 然后，学习者按照以下相同的比例准备 3 种浸泡式冲煮： - 冲煮 1-在第一 30 秒内搅拌 3 次 - 冲煮 2-连续搅拌 1 30 秒 - 冲煮 3-持续搅拌持续 30 秒 - 学习者分析每种冲煮（TDS 和感官），并注意冲煮之间的感官和可溶物浓度差异。 Part 1: Immersion turbulence comparison - Learners to prepare and analyze (in terms of TDS and sensory observations) a "Reference" immersion brew that utilizes a 1:14.5 ratio and achieves a 1.30% (+/-0.05%) soluble concentration. Learners may apply any type of suitable turbulence when preparing the reference brew. - Learners then prepare 3 immersion brews using the same ratio as follows:

- Brew 1 - Stir 3 times in 1st 30 seconds
 - Brew 2 - Stir continuously for 1st 30 seconds
 - Brew 3 - Stir continuously for last 30 seconds
- Learners analyze each brew (TDS and sensory) and note sensory and solubles concentration differences between brews.

第 2 部分：滴滤器扰流比较

- 学习者准备和分析（根据 TDS 和感官观察结果）“参照”滴滤过滤器冲煮，该冲煮过滤器利用 1:17 的比例，预润湿率为 12%至 18%，达到 1.30% (+/- 0.05) %) 可溶性浓度。在准备参照冲煮时，学习者可以应用任何类型的合适的扰流和合适的闷蒸时间。
- 然后，学习者按照相同的比例，预浸百分比和闷蒸时间准备 3 种滴滤过滤器冲煮，如下所示
(请在下面选择一个选项。)

选项 1

冲煮 1-在闷蒸期间，在所有闷蒸时间内搅拌或摇晃咖啡。根据参照冲煮剩余水量。

冲煮 2-按照参照冲煮进行闷蒸。剩余注水应尽可能快。

冲煮 3-按照参照冲煮进行闷蒸。剩余注水应尽可能缓慢而不失速。

- 学习者分析每种冲煮（TDS 和感官），并注意以下两种冲煮之间的差异：感官，接触时间，可溶物浓度和萃取率

选项 2

冲煮 1-在闷蒸期间，在所有闷蒸时间内搅拌或摇晃咖啡。根据参照冲煮剩余水量。

冲煮 2-在所有注水期间，水壶尖应尽可能靠近咖啡床。闷蒸时间应与参照冲煮时间相同。

冲煮 3-按照参照冲煮进行闷蒸。剩余注水应尽可能快。

学习者分析每种冲煮（TDS 和感官），并注意以下两种冲煮之间的差异：感官，接触时间，可溶物浓度和萃取率

Part 2: Drip filter turbulence comparison

- Learners to prepare and analyze (in terms of TDS and sensory observations) a "Reference" drip filter brew that utilizes a 1:17 ratio, with a 12% to 18% pre-wet, and achieves a 1.30% (+/-0.05%) soluble concentration. Learners may apply any type of suitable turbulence and suitable bloom time when preparing the reference brew.
- Learners then prepare 3 drip filter brews using the same ratio, pre-wet %, and bloom time as follows:
(Please choose an option below.)

Option 1

Brew 1 - During the bloom, stir or swirl the coffee for all of the bloom time. Remaining water delivery as per the reference brew.

Brew 2 - Bloom as per reference brew. Remaining water delivery to be as fast as device will allow.

Brew 3 - Bloom as per reference brew. Remaining water delivery to be as slow as possible without stalling.

- Learners analyze each brew (TDS and sensory), and note the differences between the brews in the following: sensory, contact time, solubles concentration and yield

Option 2

Brew 1 - During the bloom, stir or swirl the coffee for all of the bloom time. Remaining water delivery as per the reference brew.

Brew 2 - During all water delivery, kettle tip to be as close to coffee bed as possible. Bloom time to be the same as the reference brew.

Brew 3 - Bloom as per reference brew. Remaining water delivery to be as fast as device will allow.

- Learners analyze each brew (TDS and sensory), and note the differences between the brews in the following: sensory, contact time, solubles concentration & yield

3.03.07.07 水质测试 Water Testing	学习者 1.使用建议的工具测量各种冲煮水组合物的总硬度，碱度，pH 和电导率。 2.如有必要，转换测量结果，在 SCA 水表上绘制总硬度和碱度测量值，并评估每个样品是否在建议范围内。 3.使用“您的水是否适合冲煮工作表”，预测每种水对冲煮咖啡的感官特性和冲煮设备的可能影响。 注意：在此活动中使用的咖啡一定不能在操作考试中使用。 Learners 1. Use recommended tools to measure total hardness, alkalinity, pH, and electrical conductivity of various brewing water compositions. 2. Convert measured results if necessary, plot total hardness and alkalinity measurements on the SCA Water Chart and evaluate whether each sample is within the recommended range. 3. Using the “Is your water suitable for brewing worksheet”, predict the likely effects of each water on brewed coffee's sensory attributes and brewing equipment health. Note: The coffee used in this activity MUST NOT be used in the practical exam.
3.03.07.08 Verifying Sensory Outcomes	第 1 部分： 学习者将使用 4 种不同的水冲煮和品尝相同的咖啡和冲煮方案： • 水 1：碱度= 10 至 20 ppm CaCO₃; TH = 20 至 30 ppm CaCO₃; pH = 6.5 至 8.0; EC = 10 至 1000 µS / cm • 水 2：碱度= 40 至 60 ppm CaCO₃; TH = 50 至 100 ppm CaCO₃; pH = 6.5 至 8.0; EC = 10 至 1000 µS / cm • 水 3：碱度= 10 至 35 ppm CaCO₃; TH => 200 ppm CaCO₃; pH = 6.5 至 8.0; EC = 10 至 1000 µS / cm • 水 4：碱度=> 140 ppm CaCO₃; TH => 200 ppm CaCO₃; pH = 6.5 至 8.0; EC = 10 至 1000 µS / cm Part 1: Learners will brew and taste the same coffee and brew plan using 4 different waters: • Water 1: Alkalinity = 10 to 20 ppm CaCO₃; TH = 20 to 30 ppm CaCO₃; pH = 6.5 to 8.0; EC = 10 to 1000 µS/cm • Water 2: Alkalinity = 40 to 60 ppm CaCO₃; TH = 50 to 100 ppm CaCO₃; pH = 6.5 to 8.0; EC = 10 to 1000 µS/cm • Water 3: Alkalinity = 10 to 35 ppm CaCO₃; TH = > 200 ppm CaCO₃; pH = 6.5 to 8.0; EC = 10 to 1000 µS/cm • Water 4: Alkalinity = >140 ppm CaCO₃; TH = >200 ppm CaCO₃; pH = 6.5 to 8.0; EC = 10 to 1000 µS/cm ➤ 注意：这些规格允许高 EC 存在，前提是 EC 的增加不是由于水中存在不希望的化合物（例如 NaCl）引起的。 ➤ 为了获得最佳效果，预计咖啡会呈现出中等的口感和中等强度，然后建议使用高品质的酸度。有关硬化和碱化溶液配方的信息，请参阅《SCA 水质手册》第 12.8 节，该配方可用于为该练习创建水。 ➤ 注意：优选处理水使其达到 SCA 建议的总硬度和碱度范围，但是专业咖啡师应能够调整其冲煮方法，以在无法进行处理的情况下改善其效果。 ➤ Note: These specifications allow for a high level of EC, assuming that the increased EC is not due to undesirable compounds being present in the water (e.g. NaCl). ➤ For best results, a coffee which is expected to exhibit a medium body and moderate intensity, then high-quality acidity is recommended. Refer to section 12.8 of the SCA Water Quality Handbook for hardening and alkalinizing solution recipes which can be used to create the water for this exercise. ➤ Note: Treating water to bring it within SCA recommended ranges of total hardness and alkalinity is preferred, however the professional brewer should be able to adapt their brewing to improve their results in situations where treatment is not possible. 可能的冲煮计划更改可能包括： • 当碱度和/或总硬度较低时，增加可溶物的产量以增加口感并降低可感知的酸度。

	- 当碱度高时，减少可溶物的产量以增加可感知的酸度。 第 2 部分： - 学习者比较每种水冲煮的感官评价，然后制定计划以改善其中一种水的冲煮效果。 - 学习者准备额外的冲煮，然后分析和讨论结果。 Possible brew plan changes could include: - Increasing solubles yield to increase body and reduce perceived acidity when alkalinity and/or total hardness are low. - Reducing solubles yield to increase perceived acidity when alkalinity is high. Part 2: - Learners compare the sensory evaluations of the brews from each water and then create a plan to improve the results from 1 of the waters. - Learners prepare the additional brew, and then analyze and discuss the results.
3.03.08 过滤介质 Filter Media	学习者使用相同的滴滤器设备但使用不同的过滤介质（即纸张，金属（筛子）和布）准备多种冲煮，并比较冲煮之间的感官特性。学习者应特别注意以下方面的差异： - 口感 - 液体清彻度 - 风味清晰度 - 甜度，酸度和苦味的感知 - 学习者使用滤纸准备参照冲煮。参照冲煮的最低可溶物浓度应为 1.15%，建议的可溶物产率为 18%至 22%。 - 学习者完成参照冲煮的感官分析。 - 然后，学习者使用相同的参数从以下每种过滤介质中产生一种冲煮： - 金属（筛） - 布 - 学习者完成对冲煮的感官分析，以便与参照冲煮进行比较，并注意所有冲煮之间的接触时间和可溶物浓度的任何差异。 - 如果任一种替代过滤介质产生的溶出物浓度与参照冲煮不同（> 0.05% +/-差异），则学习者应准备一种过滤介质，以达到与参照冲煮相似的溶出物浓度（≤0.05%） +/-差），使用相同的咖啡与水的比例，但通过更改任何其他参数（例如，研磨设置，供水时间，脉动等）。 - 然后，学习者完成对冲煮物的感官分析，以实现可溶物浓度的相似性并与参照冲煮物进行比较。 Learners prepare multiple brews using an identical drip filter device, but with different filtering media (i.e. paper, metal (sieve), and cloth), and compare sensory characteristics between the brews. Learners should particularly note differences in: - Mouthfeel - Clarity of liquid - Clarity of flavor - Perception of sweetness, acidity, and bitterness

	- Learners prepare a reference brew using a paper filter. The reference brew should have a minimum solubles concentration of 1.15%, and a recommended solubles yield of 18% to 22%. - Learners complete sensory analysis of the reference brew. - Learners then use identical parameters to produce one brew from each of the following filtering media: - Metal (sieve) - Cloth - Learners complete sensory analysis of the brews for comparison with the reference brew, and note any differences in contact time, and solubles concentration between all of the brews. - If either of the alternative filtering media produces a different solubles concentration ($>0.05\%$ +/- difference) from the reference brew, learners are to prepare a brew with that filtering medium that achieves a similar solubles concentration to the reference brew ($\leq 0.05\%$ +/- difference), using the same coffee to water ratio, but by changing any other parameter/s (e.g. grind setting, water delivery duration, pulsing, etc.). Learners then complete sensory analysis of the brews that achieve similarity of solubles concentration and compare with the reference brew.
3.04.03.05 润湿应用 Applying Wetting	滴滤冲煮的闷蒸比较 学习者根据 TDS 和感官观察来准备和分析“参照”滴滤冲泡，如下所示： - 比例 1:17 - 润湿水=每克咖啡粉 2.0 至 2.5 克水（在 AFB 上预润湿 14%） - 暂停 30 秒用于闷蒸 1.30%可溶性浓度（+/- 0.05%） - 建议的可溶物萃取率为 18%至 22% 然后，学习者使用相同的咖啡与水的比例准备 5 个滴滤设备，并为每个冲煮设备更改一个变量，如下所示： - 冲煮 1-润湿水=每克咖啡粉 1.0 至 1.5 克水（在 AFB 上预润湿 10%） - 冲煮 2-润湿水=每克咖啡粉 3.5 克至 4.0 克水（在 AFB 上预润湿 18%） - 冲煮 3-在闷蒸期间对咖啡床进行搅拌或旋转 - 冲煮 4-15 30 秒用于闷蒸 - 冲煮 5-45 30 秒用于闷蒸 学习者会分析每种冲煮（TDS 和感官），并与参照冲煮进行比较。 Drip filter bloom comparison Learners prepare and analyze (in terms of TDS and sensory observations) a "Reference" drip filter brew as follows: 1:17 ratio - Water for wetting = 2.0 to 2.5g of water for each gram of ground coffee (14% pre-wet on AFB) 30 second pause for bloom 1.30% solubles concentration (+/- 0.05%) - Recommended 18% to 22% solubles yield Learners then prepare 5 drip filter brews using the same coffee to water ratio, and change one variable for each brew as follows: - Brew 1 - Water for wetting = 1.0 to 1.5g of water for each gram of ground coffee (10% pre-wet on AFB) - Brew 2 - Water for wetting = 3.5g to 4.0g of water for each gram of ground coffee (18% pre-wet on AFB) - Brew 3 - Apply stirring or swirling to the coffee bed during the bloom

	- Brew 4 - 15 second pause for bloom - Brew 5 - 45 second pause for bloom Learners analyze (TDS and sensory) each brew and compare against reference brew.
3.05.03 可溶性产率 (萃取率) Soluble Yield (Extraction)	学习者制定冲煮计划，并使用冲煮设备以相同的浓度冲煮 16%，18%，20%，22%和 24%提取率 (+/- 0.5%) 的咖啡。（建议每次冲煮的强度为 1.30%。但是，学习者可以选择适合文化偏好的能力。每种冲煮的强度应在彼此的 0.05%之内。） 学习者应该为此练习选择一种可以稳定使用的冲煮设备。例如，可以使用自动重力滴滤机。 对于每种咖啡，学习者将使用折光仪测量结果，使用咖啡冲煮控制图进行分析，并使用来自 Coffee Taster 风味轮的描述符记录感官观察结果（包括香气，味道，口感和颜色）。 在讨论感官观察时，AST 应当寻找广义上正确的描述，并提供其他相关的描述符。 完成后，AST 应当另外讨论每个学习者偏爱哪种冲煮方法以及原因。由于可以在不同强度和提取水平下找到不同咖啡的最佳平衡，因此此练习没有定义好的“正确答案”。 Learners formulate brew plans and use a brewing device to brew coffees at 16%, 18%, 20%, 22%, and 24% extraction (+/- 0.5%) at the same strength. (Recommended strength for each brew is 1.30%. However, learners may choose a strength to suit cultural preferences. The strength of each brew should be within 0.05% of each other.) Learners should select one brewing device for this exercise that they can use consistently. An automatic gravity drip brewer may be used. For each coffee, the learner is to measure results using a refractometer, analyze using the coffee brewing control chart, and record sensory observations (including aroma, taste, body, and color) using descriptors from the Coffee Taster's Flavor Wheel. When discussing sensory observations, ASTs should look for descriptions to be broadly correct and provide additional relevant descriptors. Upon completion, ASTs should additionally discuss which brew each learner preferred overall and why. Since optimum balance for different coffees may be found at different levels of strength and extraction, there is not a defined "right answer" for this exercise.
3.05.04 强度 Strength	该活动旨在通过以下方式提高学习者优化冲煮咖啡的能力： This activity is designed to increase the learner's ability to optimize a brewed coffee by: - 使用可用的咖啡，磨豆机和冲煮设备提高其增加和降低冲煮强度的能力 Improving their ability to increase and decrease brew strength using the available coffees, grinders, and brewing devices - 允许学习者收集和解释不同冲煮强度下的咖啡感官观察结果。 Allowing the learner to collect and interpret sensory observations about the coffee at different brew strengths. - 学习者应制定冲煮计划，并使用冲煮设备以相同的萃取量（例如：1.15%，1.35%和 1.55%浓度 (+/- 0.05%)）冲煮咖啡。 19%+/- 0.5%。 Learners should formulate brew plans and use a brewing device to brew coffees at 1.15%, 1.35%, and 1.55% strength (+/- 0.05%) at the same extraction, e.g. 19% +/-0.5%. - 学习者应该为此练习选择一种可以持续使用的冲煮设备。可以使用自动重力滴滤机。 Learners should select one brewing device for this exercise that they can use consistently. An automatic gravity drip brewer may be used. - 对于每种咖啡，学习者将使用折光仪测量结果，使用咖啡冲煮控制图进行分析，并使用来自 Coffee Taster 风味轮的描述符记录感官观察结果（包括香气，味道，身体和颜色）。 For each coffee, the learner is to measure results using a refractometer, analyze using the coffee brewing control chart, record sensory observations (including aroma, taste, body, and color) using descriptors from the Coffee Taster's Flavor Wheel. - 在讨论感官观察时，AST 应当寻找广义上正确的描述，并提供其他相关的描述符。 When discussing sensory observations, ASTs should look for descriptions to be broadly

	correct and provide additional relevant descriptors. 完成后，AST 应当另外讨论每个学习者偏爱哪种冲煮方法以及原因。由于可以在不同强度和萃取水平下找到不同咖啡的最佳平衡，因此此练习没有定义好的“正确答案”。Upon completion, ASTs should additionally discuss which brew each learner preferred overall and why. Since optimum balance for different coffees may be found at different levels of strength and extraction, there is not a defined "right answer" for this exercise.
3.06.01 磨豆机刀盘 Grinder Burrs	理想情况下，AST 将具有两个相同的平面或圆锥形磨豆机刀盘，如下所示： • 一台带有新刀盘或老刀盘的磨豆机 • 一台砂轮严重磨损和/或损坏的磨豆机 磨豆机不是必须相同的，因为该活动的目的是了解刀盘磨损如何影响成品咖啡。 AST 准备 2 种冲煮咖啡-一种是新刀盘研磨的咖啡粉，另一种是从旧刀盘研磨的咖啡粉。每种冲煮均在浸泡式冲煮器中进行如下准备： • 1: 15.4 咖啡与水的比例 • 达到 1.30%的可溶物浓度 (+/- 0.05%) 学习者完成对冲煮的感官分析，并确定刀盘与冲煮质量之间的关系。 注意：目的是生产在相同的咖啡与水比例下达到相似的可溶物浓度的冲煮。尽管可溶物的浓度可能相似，但冲煮质量应有显著差异。为了更有效地促进此练习，可以在过程之前准备咖啡粉样品，并将其保存在密封的袋子/容器中。 Ideally, the AST will have 2 identical flat or conical burr grinders as follows: • One grinder with new or seasoned burrs • One grinder with heavily worn and/or damaged burrs It is not essential that the grinders are identical, because the intention of the activity is to understand how burr wear affects the finished brew. AST prepares 2 brews - one brew from coffee ground with the new/seasoned burrs, and one brew from coffee ground with the heavily worn/damaged burrs. Each brew is prepared in an immersion brewer as follows: • 1:15.4 coffee to water ratio • Achieves 1.30% solubles concentration (+/- 0.05%) Learners complete sensory analysis of the brews and identify the relationship between the burrs and the quality of the brews. Note: The intention is to produce brews that achieve similar solubles concentration from the same coffee to water ratio. Although the solubles concentration may be similar, the brew quality should differ significantly. To facilitate this exercise more efficiently, both of the ground coffee samples may be prepared prior to the course and held in sealed bags/containers.
3.07.03 旁路对冲煮咖啡感官方面的影响 Effect of Bypass on the Sensory Aspects of Brewed Coffee	学习者准备 4 杯冲煮咖啡-2 杯带有滴滤设备，2 杯带有备用设备（例如浸入式）。 • 来自每个设备的第一个冲煮无需旁路即可制备，来自每个设备的第二个冲煮无需旁路即可制备。 非旁路，滴滤过滤器冲煮： • 学习者选择可以使口味和强度保持平衡的配方 • 测量并绘制冲煮图，并记录感官特征 使用旁路，滴滤过滤器冲煮： • 学习者使用 10%到 20%的旁路来计算和准备配方，与非旁路冲煮可达到相似的可溶物产量

- 分析冲煮的感官特性，并与非旁路冲煮进行比较。

非旁路，用另外的设备冲煮：

- 学习者选择可以使口味和强度保持平衡的配方
- 测量并绘制冲煮图，并记录感官特征

使用旁路，用另外的设备冲煮：

- 学习者使用 10%到 50%的旁路来计算和准备配方，与非旁路冲煮可达到相似的可溶物产量

分析冲煮的感官特性，并与非旁路冲煮进行比较。

Learners prepare 4 brews - 2 with a drip filter device, and 2 with an alternative device (e.g. Immersion).

- The first brew from each device is prepared without bypass, and the second brew from each device is prepared with bypass.

Non-bypass drip filter brew:

- Learners choose a recipe that produces a pleasant balance of taste and strength
- Measure and chart the brew, and note sensory characteristics

Bypass drip filter brew:

- Learners calculate and prepare a recipe using 10% to 20% bypass, that achieves similar solubles yield as the non-bypass brew
- Analyze the sensory characteristics of the brew and compare with the non-bypass brew.

Non-bypass alternative device brew:

- Learners choose a recipe that produces a pleasant balance of taste and strength
- Measure and chart the brew, and note sensory characteristics

Bypass alternative device brew:

- Learners calculate and prepare a recipe using 10% to 50% bypass, that achieves similar solubles yield as the non-bypass brew
- Analyze the sensory characteristics of the brew and compare with the non-bypass brew.